

CSU – RAMS

Model Update and Modification Details

This document contains detailed notes of changes made between model versions as development and deprecation were performed on the model by Stephen Saleeby and others at CSU starting in 2010. This is meant primarily as a reference for noting when particular changes take place in CSU-RAMS.

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Last updated: Aug 28, 2026

DESCRIPTION OF MODIFICATIONS MADE TO RAMS BETWEEN VERSIONS

VERSIONS ARE UPDATED WITH A DATE STAMP and/or VERSION NUMBER

Some changes were prompted by bounds checking via the compiler. You must compile with "-Mbounds" for PGI or "-check bounds" for IFORT and do a test run.

#####

Moving from Past Versions to rams6_20100504

*For the micro budgets I set the k=1 level to the k=2 level value. (file mic_driv.f90)

*Set all latent heat budgets to run over "do k=2,m1" (file mic_driv.f90)

#####

Moving from rams6_20100504 to rams6_20101004:

*In geodat.f90 added statement to solve dateline SST issues

*In landuse_input.f90 put min/max functions for glonp1 for dateline issues

*In rad_driv.f90 put min/max bounds for gp and omgp to keep them physical

*Added IBUBBLE option section to RAMSIN, condition to rhhi.f90 and specification to "bubble" in ruser.f90. This added the variables:

IBUBBLE,IBDXIA,IBDXIZ,IBDYJA,IBDYJZ,IBDZK1,IBDZK2,BTHP,BRTP

#####

Moving from rams6_20101004 to rams6_20101207:

*Added IDATAIN to specify different ingest datasets. Currently IDATAIN = 1 is for NARR and IDATAIN = 0 for all others. Would need to adjust this for other possible grids.

*Remove call to lambcon_adjust from asti.f90. Do not need this since RAMS6 apparently added a routine called "rotate_winds" to deal with rotating winds from different projections.

*Added new variable IDATAIN into the calls to "rotate_winds" in asti.f90. And in "rotate_winds" in "map_proj.f90" insert special statement whereby the winds from NARR data are not rotated to earth-relative since they are already earth-relative in the dataset. This is a special case.

*Added special IDATAIN statement for NARR data in file "astp.f90" in routine "pressure_stage" in the possible conversion of ingested specific humidity to relative humidity. The current statement assumes in input specific humidity in g/kg, but NARR is kg/kg. We may need to add more special statements for additional datasets.

*Added hybrid power law tables for hydrometeor fall speeds that use new equation for rain, pristine, graupel, hail, but original ones for snow and aggregates. (file mic_init.f90, routine micro_master)

*To fit this isan version to correct version for NARR and selectively for other datasets: In "isnstage" in asti.f90, comment out "lambcon_adjust". In "pressure_stage" in astp.f90 add situation for NARR which gives specific humidity in kg/kg, not the assumed g/kg. In "rotate_winds" in map_proj.f90 in section for (rot_type == 'lc_rps') comment out call to "uvlc_uvll" and use call to "uevetouv" as in section for (rot_type == 'll_rps').

*Changing T(kelvin) to T(celcius) by subtracting 273.15 instead of 273.16 (file kf_cuparm.f90, kf_driver.f90, oda_proc_obs.f90, rhhi.f90, rprnt.f90, asti2.f90)

*Remove density multiplier from "colface = colfacr * dn0" since density was previously factored into the equation (file mic_misc.f90, routine each_column)

#####

Moving from rams6_20101207 to rams6_20110712:

*New cloud nucleation lookup tables that add a dimension of solubility (epsilon). Now in a separate file mic_nuctab.f90.

*Fixed and updated dust source model. Appropriate for idealized runs. Fixed and updated aerosol, salt, and dust dry and wet deposition. Added salt_dust_deposit.f90, salt_dust_include.f90, salt_dust_sources.f90.

*Fixed and updated aerosol, salt, dust radiative effects. The version by Dave Stokowski had serious errors in calculation of single scatter albedo and asymmetry parameter.

*Jerry's fix to short wave flux routine in rrads.f90 can still allow the occasional division by zero error. Will go with the OLAM fix for now.

*ICCNLEV for aerosol depletion and restoration and tracking of aerosol mass in some of the hydrometeor species.

*New hydrometeor initialization routines for setting up aerosol log-normal distribution based on more recent parcel model simulations.

*Limiting enemb to only be used under certain conditions. (file mic_misc.f90, routine enemb)

- *Putting bounds on use of supersaturation tables in cloud nucleation. (files mic_nuc.f90 or mic_nuctab.f90, routine cloudnuc/aero_nuc_tab)
- *Limit amount of cloud water transfer during homogeneous ice nucleation. (routine icenuc)
- *Positive definite checks added to collection transfers for rx and cx. (routine colxfers)
- *Set upper size limit of GCCN to 5.0 microns. (routine cldnuc)
- *Fixed minor error in establishing k levels of cloud water after nucleation (cldnuc)
- *Added output of additional radiation variables, 3D precipitation rate, vertical velocity terms budgets.

#####

Moving from rams6_20110712 to rams6_20120202:

- *Separated out aerosol species into CCN, GCCN, dust1, dust2, salt file, salt jet, salt spume, sub-micron and super-micron regenerated aerosol. These are all nucleated separately in cloud nucleation.
- *Created file called mic_nucpre.f90 called prenuc_ccn to check all aerosol for number, mass, and size limits and determine the ratio of cloud water that each gets at nucleation based on total species surface area. Using revised formula to go from median radius to mean radius $\exp(1.5 * (\log(\sigma_{ccn}))^2)$
- *Installed revised DeMott formula for heterogeneous ice nucleation based on all aerosol species concentration (except sea salt) > 0.5 micron diam. Apply a pre-call to this in cloudnuc to routine prenuc_ifn in order to preferentially set aside aerosols to act as IN. Then remove those from the available aerosols for cloud nucleation.

#####

Moving from rams6_20121207 to rams6_20130409:

- *Fixed an implementation of DeMott(2010) IN and made flag IIFN = 1,2,or 3. The user can now use it like the Meyers scheme with an IN profile in CIFNP.
- *Added soil moisture, soil temperature ingest for 2 layers. Also ingest of snow water content and snow depth. For now this is just for NARR and GFS data and the user must use new degribber code to extract these data. These fields are added to the varfiles and then ingested appropriately at the model start. Routines to set up the soil and snow fields are now in leaf_init.f90.
- *Made a couple minor changes to rsys.F90 and rammain.f90 for compiler and system commands. This changes allows RAMS to compile on standard Linux and Mac Linux with PGI, IFORT, or GFORTRAN.
- *In sfc_driver.f90 added calls to new boundary condition subroutines. Added call to "turb_bcond" for sflux_u, sflux_v, sflux_w, sflux_r, sflux_t and added call to "rad_bcond" for rlongup and albedt. These two routines have been added after "leaf_bcond" in the file leaf3.f90

#####

Moving from rams6_20130409 to rams6_20130813:

Mods to Powers Laws and Sedimentation

- *Added True-Bin sedimentation in addition to the Standard version using displacement distance and "reldisp". I have kept both for now, but perhaps the old version could be removed in the future. This new Sedimentation is based on work from Adrian Loftus.

*"nembfall" in mic_driv.f90 was increased from 20 to 40 to better cover mean mass parameter space

*"nbins" was set to 41 in mic_tabs.f90. Bins are mass doubled and span the range of allowable mean masses.

*In the setup of gamma distributions across bins, the lowest and highest 5% of bins are removed to prevent including and large tail of mass and the small tail of number. Distributions are then normalized. Note that this 5% should be carefully changed if "nbins" is changed in the future.

*Added Rob Carver's power law bands to micphys.f90 and sedimentation lookup tables (mic_tabs.f90,mic_misc.f90,mic_driv.f90).

*Added dimensions of "ndensrtgt" and "nband" to sedimentation lookup tables. "nband=3" is for implementation of Rob Carvers banded power laws. This works only in the True-Bin sedimentation. "ndensrtgt=40" dimension covers parameter space for the density and rtgt effects on fall distance. See tables below:

```
pcp_tab(ngr)%pcpfillc(mmzp(ngr),maxkfall,nembfall,nhcat,ndensrtgt,nband))
pcp_tab(ngr)%pcpfillr(mmzp(ngr),maxkfall,nembfall,nhcat,ndensrtgt,nband))
pcp_tab(ngr)%sfcpcp(maxkfall,nembfall,nhcat,ndensrtgt,nband))
pcp_tab(ngr)%allpcp(mmzp(ngr),nembfall,nhcat,ndensrtgt,nband))
```

*In true-bin sedimentation we now bilinearly interpolate for mean mass and power law band lookup tables dimensions to remove noise that exists otherwise

*IPLAWS in RAMSIN: 0-Original4.3, 1-New Single Choice (R.Carver,Mitchell96) 2-New With Banded Plaws for sedimentation (R.Carver,Mitchell96) A value of 2 only works for new True-bin sedimentation

*ISEDIM added to RAMSIN: Sedimentation: 0-RelDisp, 1-True Hybrid-Bin

*In mic_driv.f90 has to add option of setting of "ch2" variable for mean mass rather than displacement distance. Also added new option for "mksedim_tab" called "mksedim_tab_trubin" and new option of "sedim" called "sedim_trubin".

*Removed mass and Vt power law table "dstprms3" in mic_init.f90. The banded power laws should replace this option.

Other Modifications

*Bug fix to "nucclدت" micro budget variable. In former versions this was possibly incorrect when ICCNLEV=0. Added separate calculations for the case of ICCNLEV=0 and for ICCNLEV>0. (Mods to mic_nuc.f90)

*Budget variables dust1cldrt, dust1drzrt, dust2cldrt, dust2drzrt are now only calculated for ICCNLEV>0 as they were incorrectly computed for ICCNLEV=0. The code would have to be greatly altered to have these budgets available for ICCNLEV=0. (Mods to mic_nuc.f90 and opspect.f90)

*In mic_tabs.f90 subroutine sxy, added line to keep number and mass bins consistent: "amkd(1)=0. !If bin number is zero, make sure mass is zero."

#####

Moving from rams6_20130813 to rams6_20130901:

*Added cloud droplet nucleation lookup table for NaCl. Renamed the routines for accessing the lookup tables to delineate between (NH4)2SO4 and NaCl.

*In micphys.f90, added an array to specify soluble chemistry.

*In mic_init.f90 added a routine (aerosol_init) to set the vanthoff factor, (really the number of dissociated ions in solution) to the soluble chemistry. The vanthoff factor is used in (cal_dwet). This routine also determines the dry aerosol density based on weighted densities of the solid and soluble materials determined by the solubility fraction. The call to this routine is added for initial and history restarts in rdint.f90.

*In aerosol deposition routine, changed the condition for the call to cal_dwet (deliquescence growth routine) from vanthoff factor related to solubility fraction and RH related.

*In cal_dwet also inserted code to allow for different aerosol dry densities in deliquescence equation. Also inserted code to compute the density of the deliquesced solution particle to coincide with radius of the wet particles when used in wet and dry deposition routines. More deliquesced particles should have net density closer to that of water (1000 kg/m3).

*Fixed bugs in cal_dwet for a mix of radius units in meters and microns. Set an upper limit for wet particle sizes to 10 microns radius. Larger particles should not be allowed for gravitational settling for the equation being used since it is only appropriate for falling particles with low Reynold's numbers $< \sim 0.1$.

*In mic_init.f90 I moved the call to "prenuc_ccn" outside the supersaturation requirement so that it does an aerosol size computation and size limitation for all grid cells. Turns out this is necessary to keep particles sizes in bounds where nucleation is not occurring, but advection and diffusion and mess with the computed sizes.

#####

Moving from rams6_20130901 to rams6_20131005:

*Reorganized some code structure to make things run more efficiently when NOT using ICCNLEV $\geq$ 2. Just set things up so there's no unnecessary "if" checks or "do" loops.

*Got rid of ICCNLEV==3 option. Rather I introduced three flags to set in micphys.f90. These are "trackdust", "trackdustifn", and "trackepsilon". (trackdust=1 allows tracking of dust within hydrometeors and allocates 8 new arrays.) (trackdustifn=1 allows tracking of dust that enters hydrometeors from ice nucleation only and it allocates 8 new arrays.) (trackepsilon=1 allows tracking of aerosol solubility as a whole and allows prediction of epsilon (solubility fraction) for the regenerated aerosol categories. This requires 8 tracking arrays and 2 arrays that hold the amount of soluble regenerated aerosol mass. We then get epsilon from the ratio of soluble/total aerosol mass.

*Fixed a limited in IIFN=3 for DeMott nucleation. Was nucleating new ice like "nifn = nifn - (total_in + immersed + ifnnuc)" but should be like "nifn = nifn - ifnnuc". Fixed this in mic_nuc.f90 & mic_nucpre.f90 and updated total_in when particles enter as potential immersion freezing nuclei.

#####

Moving from rams6_20131005 to rams6_20131008:

*Included bug fix found by Adele Igel to limit lapse rates in turb_diff.f90 subroutine truhor. Indices were improper.

*Throughout the model, I converted all aerosol number and mass concentrations from #/cm3 and g/cm3 over to #/kg and kg/kg. Did this so that aerosol fields are conserved during advection, since advection does not currently conserve volume fields.

*Decided to remove nucldct and nucicect from budgets. The number concentration can get very large when accumulated and not sure how useful these are compared to the mixing ratios which

come from conserved quantities. May remove other number concentration budgets as well, but not in this release.

#####

Moving from rams6_20131008 to rams6_20131111:

- *Removed all the print out and old rams plotting sections from RAMSIN and from the code. This is the start of a cleanup effort to remove obsolete and unnecessary portions of the code.
- *Removed microphysical budgets for nucleation number concentration
- *Created lateral boundary points for radiation variables since they formerly were simply set to zero. This created odd looking gradients at all boundaries when plotting, and could impact analyzed results.
- *Reorganized the namelist info in rcio.f90, rname.f90, and mpass_init.f90 to make it easier to add variables in the future and keep them ordered.
- *Commented out a lot of print statements just to clean up the standard output for statements we really never use. Can turn these back on if we need to.
- *Got rid of calls to "numcheck" in the micro driver. These are no longer needed for IIFN=3 since we run this check in the negative adjustment call.
- *Deleted references to LEVEL = 4 and 5 microphysics which used to be a bin sub-model, but the code was never present in standard RAMS. Plus, we will have the HUCM bin model in future version of CSU-RAMS and will not need these old options.
- *In rams_grid.f90 subroutine newgrid, I broke up a long do loop into several smaller do loops to allow the IFORT compiler to vectorize this loop more quickly at compile time for -O2. Otherwise it is really slow to compile (many hours).
- *Reordered and added model checks in opspec.f90.
- *In varf_update.f90 changed the varfile variable copying (future to past) from implicit do loops to explicit do loops. I had sometimes been getting segmentation faults at this point in the code when using the INTEL fortran compiler.
- *Added the rtgt factor that influences vertical spacing over topography, to the aerosol deposition routine, and input ubmin for minimum wind speed similar to what is done in leaf3 because errors occur is wind speed becomes zero.
- *Included a bounds fix check in "psxfer". Occasionally snow mixing ratio was becoming negative because of a computation precision difference. This produced a negative condensate error. We now limit ice-snow transfers to positive definite values.
- *Moved call to "aerosol_init" in rdint.f90 to the top of the initialization routine, so that this only has to be placed once in the code. Added passing of aero_rhosol and aero_vanthoff arrays to mpass_init.f90 to make this possible for parallel runs. Otherwise I'd have to repeat the call to aerosol_init within the timestep routine AEROSOLS.
- *Altered code to allow aerosol deposition for 1 or 2 moment micro and for moisture LEVEL = 0, 1, 2, or 3, and aerosols can be allocated and initialized for any moisture level. This is useful to look at dry deposition without micro being turned on. Also aerosol radiation can be used for LEVEL >= 1. (Radiation in general requires LEVEL >= 1.) Note that aerosols can be initialized in the model for any microphysics level and can be treated independent of the microphysics. This is useful to see how they are transported and diffused without being active. Then the user can choose the levels of activity (ie. radiation, deposition, microphysics).
- *Put call to routine "AEROSOLS" in rtimh.f90 Timestep routine. Placed this ahead of call to RADIATE. This call will run the dust and salt source models if those are active. Also will run

aerosol deposition for LEVEL<=2 if active. This replaces the call to dust and salt that was formerly in the micro driver. There is still a call to aerosol deposition in the micro driver for LEVEL==3.

- *Created temp arrays for all micro budgets that are assigned in mic_misc.f90 under range_check and are copied back in mic_driv. This solves a problem of passing arrays that may not be declared.

- *Bounds check fix to the ccncon and ccnmas array in cloud nucleation for aerosol removal.

- *Bounds check fix in auto_accret and auto_accret_ice for several variables to access collection array.

- *Found and fixed a bounds check error in rrad.f90 in the event that air temperature < 180K.

- *Fixed bug in leaf3.f90 whereby surface top level temperature was not getting initialized if radiation and soil model were turned off. This allowed canopy temperature of water patches to approach absolute zero.

- *Added RAMSIN flag ICHECKMIC which checks for negative condensate and Nans. If turned on, this calls routine “checkmicro”, which is called after MICRO in rtimh.f90.

- *Added RAMSIN flag IAEROPRNT which turns on or off printing out of initial aerosol profiles for a set I,J location.

- *Added RAMSIN flag IAERODEP to turn on and off aerosol wet and dry deposition.

- *Added RAMSIN namelist arrays for aerosol chemistry type, initial median radius, and solubility and removed these from being hard coded in micphys.f90.

- *Added RAMSIN flag IAEROHIST to allow user to reinitialize aerosol concentration profiles on a history restart and addition of new grid at history restart. This is a useful option for aerosol sensitivity tests that are run from history restart after model spinup time. Code for initialization was added in rdint.f90.

- *Added RAMSIN flag IPRNTSTMT to allow user to turn key print statement on and off for the standard output. These statements can be useful for debugging, but are clutter when trying to examine useful info in the standard output.

- *In RAMSIN – ICLLOUD, IDRIZ, IPRIS will no longer be > 5. Setting them to 5 will allow them to be 2-moment, but these are not tied to aerosol profile setup any longer. There are separate flags for these. The flag IAEROSOL in RAMSIN is set to 0 or 1, with 1 being ON. If on, then aerosol will be initialized regardless of microphysics level. Same for IDUST and ISALT.

- *In RAMSIN – PPARM should now be always 0. This is not used. There is a separate variable CIN_MAX to put the max ice nuclei variable. This is similar to how this is done for CCN_MAX, GCCN_MAX, DUST1_MAX, etc.

- *Default aerosol profiles are now set to decrease similarly to air pressure. The user needs to be aware of this and set their profiles as needed in mic_init.f90.

#####

Moving from rams6_20131111 to rams6_20131130:

- *Removed all code in RAMS related to NCAR graphics and removed the NCAR graphics from compile-time include.mk.opt and utils compile directory. Removed ncarg_dummy.f90, rams_wgrib.c, wgrib.c from utils directories.

- *Removed interface.h since it is obsolete. Removed all “include interface.h” lines of code from all routines in the model.

- *Removed a number of obsolete routines that are in the code but never call and used.

- *Removed large sections of code that have been commented out since early RAMS 4.3.

- *Added a few more print statements under the conditions that IPRNTSTMT=1.
- *In mpi/par_decomp.f90 changed the “relspeed” array dimensions from 256 to 2560. This value represents the maximum number of parallel processors on which the model can run.
- *Cleaned up urban canopy code a bit. Not sure how functional this code is in its current form.
- *In core/rammain.f90, increased the dimension of “taskid” from 512 to 2560. As with “relspeed”, this array size represents the maximum number of processor cores the model can use. This number made need to be increased again in the future.
- *In memory/grid_dims.f90, changed “MAXMACH” from 512 to 1024. This is the maximum number of processor cores the model can run on. Memory is set aside for arrays containing dimension MAXMACH. This number is limited on 32-bit systems.

#####

Moving from rams6_20131130 to rams6_20131204:

VERSION 6.0.00 (will also start using version numbers in addition to date stamps)

- *Added several more IPRNTSTMT=1 conditions for printing info to the standard output.
- *Changed dimensions of “taskids” in rammain.f90 to “maxmach”, as well as to the variables “relspeed” in par_decomp.f90.
- *Removed more obsolete or undeveloped code in the utils directories. Removed everything related to grib format and netcdf format. None of this was working. Perhaps was Tremback’s work in progress, but non-functional. Deleted from utils: fenvgen.F90, gdf_write.f90, gdf_write_sfc.f90, gdf_write_upa.f90, cdf_utils.F90, grib_c.c, grib_utils.f90, lamcon.f90, project.f90, rams_read_header.f90. None of these were compiled or used! Also deleted inthisvfm.f90 which was not compiled or used.
- *Deleted vformat.f90. This required removing code from utils and model files related to the old .vfm file format. Deleted calls to subroutines vfinal, vctran, vcorec, vcirec, vforec, vwrt, vfirec, cscale, cfact, viorec, vwrti, viirec, cscalei. Removed this file from compilation objects.mk files.
- *Deleted getvar.F90 and error_mess.f90. These were used together and only getvar.F90 called the routine in error_mess.f90. RAMS_getvar in getvar.F90 was only called by subroutines “recycle” and “first_RAMs”. It appears that “first_RAMs” is never used unless you compile with RAMS, GRIB, or WRF support. To my knowledge, no one has ever used this function which is turned on if GUESS1st=’RAMS’ within RAMSIN. Even then, I think the code needed substantial work to be functional. This could be added back in, in the future if necessary. For now, we delete first_rams.f90. We remove getvar.F90, error_mess.f90, first_rams.f90 from compilation in object.mk files within /bin and bin/utils directories.
- *Removing “first_RAMs” allows us to remove “GUESS1ST” from RAMSIN and all related namelist files. Removed reference to GUESS1ST in isan/asgen.f90, isan/asti.f90, isan/avarf.f90, isan/file_inv.f90, isan/isan_coms.f90, isan/isan_name.f90.
- *Removed obsolete and unused subroutines in multiple files in the utils/2.4 directory files. This includes cleaning up charutils.f90, dateutils.f90, interp_lib.f90, numutils.f90, parlib.c, polarst.f90, rnamel.f90, utils_f.f90, and utils_c.c.
- *Moved an_header.f90 and rconstants.f90 to the “memory” directory. From an_header.f90, I removed the subroutines “rams_read_header” and “an_setvar_info” since these were only used in getvar.F90 and recycle.f90. Further “rams_read_header” is archaic and we have better methods to do this throughout the code as is done in nud_update.f90 to read the header file.
- *Went through the code and eliminated all reference to the old .vfm file format. Reworked code sections so that hdf5 is the only i/o format. This should make the code clearer and cleaner to

work with. (Files affected were `landuse_input.f90`, `geodat.f90`, and `varf_update.f90`; also files `cu_read.f90` and `cond_update.f90` which are a special case since the code here is not functional and needs work; also `inithis.f90`, `nud_read.f90`, and `nud_update.f90` for history file initialization and nudging are discussed more in the section below. There are still references to `.vfm` in non-functional code in the routines `cu_read.f90` and `cond_update.f90`. This will be altered later.

*Added comments in `fdda/cond_update.f90`, `io/inithis.f90`, and `io/recycle.f90` that mention these features are not fully functional. The model will stop. This will be made to work in the future.

*In `rtimh.f90` added call to new subroutine “`non_scalar_bc`” at the end of `core/rams_master`. I added this routine to combine all the separate calls to set boundary grid points to non-scalar, non-advected variables. This includes calls to `turb_bcond`, `leaf_bcond`, `rad_bcond`, `sfcrad_bcond`, and `THERMO` for only BC’s. This new call to `THERMO` replaces the call that occurs just before analysis file write in `core/rams_master.f90`. We call “`non_scalar_bc`” in `core/rams_master.f90` where `THERMO` used to be called directly. Now we set all non-scalar BC’s in this location.

*Placed a fix in the vapor flux routines to prevent vapor “`rv`” from becoming negative. Tests indicate that this can sometimes occur at upper levels where vapor magnitudes are quite small. In `mic_vap.f90` in call to “`vapdiff`”, alter the following line to keep “`rv`” positive.

`rv(k) = max(0.,(rvstr(k) + sumuy(k)) / (1.0 + sumuz(k)))`

then in “`vapflux`”, altered the following line to “`rv`” positive

`rv(k) = max(0.,(rvstr(k) + sumuy(k) + sumvr(k)) / (1.0 + sumuz(k)))`

*Placed a fix in `mic_nuc.f90` in “`cldnuc`” to set aerosol concentration and mass to zero if either becomes zero. This was added in the section where aerosols are removed after nucleation.

*In `mic_driv.f90`, altered the passing of variables to “`psxfer`” so that `k1,k2` are passed for both snow and pristine ice. This change goes along with a change in `mic_vap.f90` for “`psxfer`”. The `k1,k2` values are passed in, then the min/max are used for transferring pristine ice or snow back and forth as needed. However, an error could occur such that NEW snow could be added to the snow category or vice versa without updating the `k1,k2` values resulting non-conservations of condensate mass and associated aerosol mass. So, now `k1,k2` values are updated and returned.

*Included an additional bounds check in `mic_vap.f90` to “`psxfer`” variables `dvap` and `dnum` to make sure they are always of the same sign and non-zero. I included a check statement for this and removed old statements related to “`xfer exceeded`”.

*Removed all reference to subroutine “`hydro`” and “`hydrol`” which were incomplete routines related to subsurface water tables. Call to these routines initiated from `sfc_driver.f90`. Removed the file `leaf3_hyd.f90` and removed its name from `objects.mk` for compilation. Also removed all reference to the urban canopy model, which was also an incomplete project from `ATMET`. Our `TEB` model will likely replace this urban model anyway. Removed the `RAMSIN` flag “`IF_URBAN_CANOPY`”. Removed all reference to variable “`cdrag`” for urban drag coefficients.

*Decided to remove everything related to cumulus inversion. This is still experimental and required unavailable offline code to convert observed precipitation into some sort of convective heating profile. Removed file `cuparm/cu_read.f90`, subroutine `rconv.f90/cu_inv_tend`. Also deleted all references to variables (`maxcufiles`, `maxcugrids`, `cu_times`, `ncufiles`, `fnames_cu`, `itotdate_cu`, `ncufl`, `cutime1`, `cutime2`). From `RAMSIN` and model, deleted flags (`IF_CUINV`, `CU_PREFIX`, `TNUDCU`, `WT_CU_GRID`, `TCU_BEG`, `TCU_END`, `CU_TEL`, `CU_TIL`).

*In `surface/ruser.f90`, subroutine “`eng_params`”, I removed all reference to obsolete variables named “`IZFLAT`”, “`NTOPSMTH`”, “`IADVL`”, “`IADVFL`”. Also removed routines and/or calls to routines “`toposmth`” and “`toposmooth`” as these were not used and may need additional work.

- *In varf_update.f90 changed the varfile computation of “varpf” from implicit do loops to explicit do loops. I had sometimes been getting segmentation faults at this point in the code when using the INTEL fortran compiler. Seg faults did not occur with PGF90.
- *Added GNU license header to every program file.
- *Added condition such that soil moisture and temperature and snow data are only attempted to be ingested for an INITIAL run and not for HISTORY. Also added a check to see if varfile for particular grid is available. If not, perform default soil moisture initialization as usual.
- *In mpi/mpass_feed.f90, added a print statement condition and removed subroutine “prtout”.

For making the HISTORY initialization and nudging code functional:

- *In init/rdint.f90 routine “initlz” for initialization, I started making modifications for making the HISTORY initialization option work. This will require more work to be fully functional.
- *In io/inithis.f90 made many changes that read/interpolate a previous history file into initial conditions for a new run that can be on a similar grid or a different grid. Changes thus far allow the new initial output file to equal the history file that was read in. More work needs to be done.
- *In lib/interp_lib.f90 and fdda/nud_analysis.f90 Adele make a code update to allow history initialization and nudging for simulations run in 2D.
- *Changes to fdda/nud_read.f90 to allow reading of hdf5 history files.
- *Complete redo of fdda/nud_update.f90 for nudging with history files. I do not think this this ever functioned, so it required a lot of experimentation and testing to make it work.

#####

Moving from rams6_20131204 to rams6_20140114:

VERSION 6.0.01

- *Installed Adele Igel’s change to the advection code to include a monotonic flux limiter for vertical advection and improved positive definite scheme. This is currently functional in vertical direction only. Code is available for 3D flux limiter but will require a few modifications to function best in parallel. Requires message passing an additional layer of data along boundaries. Using the 3D version without such message passing changes is still mostly fine, but not exactly binary reproducible if you change the number of running nodes.
- *Removed the following unused variables from the model related to the old TOPMODEL code and Leaf-3: sporo, ssand, sclay, sorgan, root, cmin, corg, cwat, cair, cka, ckw, romin, roorg, slcond, wtroot, tmin, xmin, ratio, romean. A number of these were in the analysis output header files and will no longer be there starting with this version. This requires an update to rcio.f90 in the model and revu.

#####

Moving from rams6_20140114 to rams6_20140119:

VERSION 6.0.02

- *Created offline script to search for subroutine and functions that are present but never called. Used this to begin removing unused subroutines and function.
- *Removing more old code and standardized some code formatting for easy “grep” of certain words such as “module”, “subroutine”, “function”, “interface”, and “entry”. Avoid using these

key words when making comments so these are set aside for easy searches. Also, end every subroutine with “end subroutine”, end every function with “end function”, end every module with “end module”, and end every interface with “end interface”.

*Removed moving grids file nest_move.f90 including all subroutines therein.

*Removed subroutines acctimes1, acctimes2, kf_eta_init, ll_lc2, uvll_uvle2, staprt, nest_interpolated_topo, prtlev, exstbuffs, mkstbuffs, dnswt_off, trtend, tridiff2, tridiff2orig, friclyr, gdf_missing_sfc, ifirstchar, isnumber, is_integer, uvll_uvle, trueps60_ps_rot, trueps60_uevetouv, trueps60_uvtoueve, ps_ps, ps_ps_rot, ismax, rsifp, rslfp, aminmax, rowcolumn, ramran, cio_f8, fillsrc_den. Some of these routines are lambert conformal grid projection routines that could be useful later. Can be accessed from original files.

*Removed RAMSIN variables “GRIDU” and “GRIDV” used for moving grids. Removed these from everywhere in the code. Also removed variables “dimove” and “djmove” from memory directory files.

#####

Moving from rams6_20140119 to rams6_20140122:

VERSION 6.0.03

*Removed ADAP cut cell code since this was only partly function and has a long way to go to be functional. If you need this vertical coordinate, you will need to move to the OLAM model.

*Removed ADAP variables lpu, lpv, lpw, volt, volu, volv, aru, arv, arw.

*Removed IF_ADAP from RAMSIN and all its dependencies.

*Removed all ADAP fortran files and subroutines that are typically named “something_adap”.

*Had to remove K level offsets due to lpw, lpu, and lpv.

*Also removed routines lpuvw_init, ctrlvols.

*Removed “entry” statement pr_hystatic_t and gave it its own subroutine.

*Removed functions “os”, “tsa”, “tw”, “esat”, “w”, and “tmr”. It is not immediately clear what these were intended for. There is no info on these functions.

*Cleaned up the parlibs.c somewhat.

*Standardized the code with the following characteristics:

1. “implicit none” used in ALL modules, interfaces, functions, and subroutine in fortran. This is written beginning in first column.
2. Subroutines are headed by syntax such as: “subroutine steve ()”. Spaces are included and this starts in first column. Subroutines end with “End Subroutine”.
3. Functions are headed by syntax such as: “integer function steve ()”. Spaces are included and this starts in first column. Functions end with “End Function steve”.
4. Modules are headed by syntax such as: “module steve ”. Spaces are included and this starts in first column. Modules end with “End Module steve”.
5. Interface statements are headed as: “interface”. This starts in first column. Interfaces end with “End Interface”.

#####

Moving from rams6_20140122 to rams6_20140126:

VERSION 6.0.04

*Removed global RAMS setup since this was never really functional. This required removing nesting/hemi.f90.

- *Removed from mem_grid.f90: nhemgrd2, nhemt, nhemu, nhemv, and all variables with dimension “maxhp”.
- *Removed NESTZ2 and NSTRATZ2 from RAMSIN and all code since these refer to the 2nd hemispheric grid that we have removed.
- *Removed global dimension MAXHP.
- *Changed all instances of NESTZ1 and NSTRATZ1 to NESTZ and NSTRATZ, this included changing related variables as well.
- *Added Meesters et al. (2008) topographic gradient surface flux fix.
- *Removed a few additional deprecated declared variables: nrzflg, lev4bins, lev5bins.
- *In core/raco.f90 in subroutine prdctv, put all contents within the “jdim” if statement for 2D simulations so that V is not predicted at all in 2D simulations.
- *In init/rhhi.f90 in subroutine arrsnd, I include a statement that sets input sounding V-winds to zero in the case of a 2D simulation.
- *In isan/astp.f90 I made a few modifications for ingesting global lat/lon datasets since it was not accepting certain datasets due to missing grid navigation. This was fixed in the dataprep code and modified here accordingly.

#####

Moving from rams6_20140126 to rams6_20140130:

VERSION 6.0.05

- *Reworked the compilation Makefile routines to combine 4 short files into 1 that the user needs to modify. Also combined the RAMS utilities directories and compilation into the full model compilation to eliminate this unnecessary extra step at compile also. It also helps to centralize the source code.
- *In testing this version I stumbled upon random segmentation faults. In some instances, the computer stack overflows. So, random looking segmentation faults are sometimes not real. You can avoid this issue by increase the stack size. On Linux type the command “ulimit –s unlimited”. This HAS to be done to run OLAM or it seg-faults on the first timestep. I am wondering now if this was been the source of random seg-faults in all versions of RAMS since the early days of 4.3. We used to get random seg-faults when create surface or varfiles. It is possible these were not really code errors, but rather, stack overflow. This is being noted in the user guide. This issue could be resolved with use of allocate/deallocate within subroutines that currently declare many local 3D variables on the stack.
- *Changed the HDF5 calls to H5Dopen and H5Dcreate to the HDF5 version 1.8 API format.
- *Fixed bounds check issue in leaf3_init.f90 for the soil data ingest section. It was looking for varfiles in simulations where varfiles do not exist.
- *In mic_nuc.f90 subroutine cldnuc, a statement was input to force cloud and drizzle droplets to nucleate at their smallest allowed size.
- *Removed everything related to addition of new scalars via RAMSIN flag NADDSC. This included removing the file memory/mem_scalar.f90 and all calls to added scalars via “scalar_g” and “scalarm_g” arrays in memory/alloc.f90. In memory/mem_tend.f90, removed several loops with reference to additional scalars and tendencies of “scalar_g” and “scalarm_g” arrays.
- *Removed “proc_type” equals 3 and subroutine “dealloc_all” as these were related to dynamic load balancing which is a non-functional feature that was removed previously.
- *Removed call to THERMO in io/ranlavg.f90 and replaced with call to “non_scalar_bc” since this seems to do a better job of setting boundary conditions for non-scalars.

*From rcio.f90/commio: Removed many variables that do not appear necessary to be sent and recorded in the analysis header file. I retained the variables necessary for history restarts and REVU. Others were removed, but I added GNU and IAEROCHEM for use in future REVU. User can do a diff command on new and original files to see the changes.

*From history_start.f90/history_start: Removed header files read-in of many variables that are not necessary for history restart. Only retained the needed ones. User can do a diff on new and original files to see the changes.

*Removed use of lambert-conformal projection. This feature was non-functional and was not recommended by ATMET to use at this point in time. Removed option IHTRAN=2, call to "grdspc_lc" from init/rams_grid.f90. Removed file init/llc_utils.f90 and its compilation in bin/objects.mk. Removed all instances of RAMSIN variables "stdlat1" and "stdlat2". Removed "lc_ll" in lib/map_proj.f90 as it is unused.

*Fixed a limitation in the Kain-Fritsch cumulus parameterization that did not allow for more than 50 vertical model levels. Set the new default to 500 levels. Ideally this code would be substantially reworked to be more inline with RAMS memory and allocation structure.

*Removed variables "iinput", "iopunt", "iversion", "runtype_save", "naddsc" from the model.

*Changes every instance of "runtype" to "trim(runtype)" to prevent future conflicts.

*Removed RAMSIN and model variables XLITE, YLITE, ZLITE as these were never used.

*Removed the ":hist:" vtable type flag from usage in the model. This is mostly obsolete since we do not have separate history and analysis files. The analysis files are the history restart files.

*Removed routine and call to "filltab_scratch" as nothing is done for this.

*Removed passing "rmin", "rmax" thru MPI as it does not appear at all necessary.

*In mic_init.f90 where collection tables are created, I added a statement that forces the read-in of the collection table, even if it is being created currently. This fixes a number truncation difference that arises between creating table values in-line and reading them from a file. These differences lead to different solutions and prevent binary duplication.

VERSION 6.0.06

*Changed syntax of call to subroutines to always begin with "CALL". This is done such for ease of case sensitive grepping and code tracking. A call should be like "CALL name (variables)" with spaces included.

*Removed subroutine "wet_scavenge_chate" and its commented out call, as it was never used.

VERSION 6.0.07

*Got "shdf5_info" working for history restarts. This just provides information on arrays and dimensions, but could also be quite useful in future development.

*Since we cannot test other machine dependent types, I have removed the "#if defined" statements to all types but PC_LINUX1, which is a standard Linux OS and is the primary system of choice. I combined all system dependent routines into rsys.F90 so that there is only 1 fortran file to modify for additional systems in the future. The C file utils_c.c and utils_sub_names.h would also need to be modified for additional machine types. But these are the only 3 files to modify.

*Modified many string lengths from 80 to 128 characters so these do not get so easily truncated when having to put long path lengths into RAMSIN. Otherwise model would crash and say that it could not find a particular file.

- *In gridset.f90 I removed reference to variables “ngra” and “ngrb” and their condition statements, as these were tied to the old non-functional moving grids.
- *Checked and reorganized all function calls to that are easily traceable. Renamed a few of them so we can grep for them more easily. Set all subroutine and function names and calls to lowercase for easier searching and matching. Made sure every subroutine that uses and function declares it as an “external”.
- *Removed function “LC” as it is never used and removed routine “dnswt” as it is never used.
- *Removed call to “deallocate” routines for the global pointer variables such as “dealloc_micro” since these are never used, and system will clear memory if model crashes or finishes.
- *Set all subroutine and function names and calls to include () at the end even if not variables are passed. This is just done for consistent grepping and searching.
- *Set up the subroutine mass_flux_bc to be functional in the model for sequential simulations. This is off by default but can be turned on at the bottom of rtimh.f90.
- *Turned entry statements “azero” and “azero2” into separate subroutines to get rid of entry statements.

#####

Moving from rams6_20140130 to rams6_20140410:

VERSION 6.1.00

- *No changes here. Just freezing this version of RAMS after a substantial amount of work, bug fixes, modifications, science changes, structural changes, and code deprecation. This will move us from versions 6.0 to 6.1 and forward.

#####

Moving from rams6_20140228 to rams6_20140410:

VERSION 6.1.01 (Merged REVU back into RAMS source code to share files)

- *Added a few variables need to merge the REVU code with RAMS including “anal_table” and “maxrevu”.
- *Comment out additional call to “H5close” that interrupts REVU functionality but does not really seem necessary for RAMS.
- *Fixed the output of the “deltazn” variable in the analysis header files.

VERSION 6.1.02

- *Used gfortran compiler with –Wall option to turn on all compiler warnings. Then used this output to eliminate from the code all unused but declared variables. Also eliminated all potentially uninitialized variables.

VERSION 6.1.03

- *Fixed reading in of isentropic/sigma-z variable initialization files in the event that these are output prior to creation of the actual variable initialization files or “varfiles”. This would be most useful for development purposes where the user needs to examine the output created during the isentropic stage of the processing of pressure level grids and conversion to RAMS isentropic and sigma-z levels.
- *Also fixed a pointer initialization in mpi/mpass_full.f90.

VERSION 6.1.04

*Made a few alterations to subroutine `non_scalar_bc` to allow it to be called from master and node and pass the correct local indices. This required passing in the used indices.

*In `core/model.f90` subroutine `par_model`, changed the use of “`nmach`” to “`nmachs`” in all calls to `par_ready` subroutine. Made changes to allow “`model`” and “`par_model`” to have a more similar structure that makes them easier to compare. This required changing the location of the time update calculation and relocating the call to compute the means in subroutine `anlavg`. Added passing of local domain grid point array sizes.

*In `memory/alloc.f90`, I made modifications to allow declaring 2D grid characteristic arrays so that these can be written to the MEAN files. For example, we really need TOPT in the MEAN files for computing many of the variables in the REVU post-processor.

*In `io/anal_write.f90` subroutine `anal_write`, made a few fixes to output the correct fields for MEAN state files and BOTH files.

*In `mpi/mpass_full.f90` subroutine `node_sendanl`, added print statement to see passed variables, and in subroutine `master_getanl`, added a condition statement to update number of received variables for MEAN state files.

*In `io/ranlavg.f90` subroutine `anlavg`, added print statement to see list of averaged variables. Modified call to subroutine `non_scalar_bc` to work on parallel nodes. This required passing in the local node array sizes rather than absolute domain dimensions.

*In `mpi/rnode.f90` subroutine `rams_node`, I moved around call location to subroutine `anlavg` and added the appropriate passing variables associated with subdomain sizes.

VERSION 6.1.05

*Cleaned up the passing of unnecessary variables among the routines that read in the RAMSIN namelist.

*Eliminated all remaining common blocks.

*Fixed an error in the history restart read of NDVI values.

*Fixed a runtime and history restart error related to the Chen/Cotton radiation routine that improperly assigned K levels regardless of topography.

*Further streamlined REVU including make the syntax for subroutine and calls conform to that now used in RAMS.

VERSION 6.1.06

*Made a fix so that dust-in-hydrometeor variables are not declared and used unless both `IDUST>0` and dust tracking flags are turned on.

*Transferred a REVU file inventory “`MAX`” variable from `rcommons.f90` to `grid_dims.f90`.

*Made a fix to collection of cloud droplets. In past versions, lookup tables in subroutine “`make_autotab`” in file `mic_tabs.f90` started with a minimum mixing ratio of 1.e-5 kg/kg while the application of these lookup tables started at 1.e-12 kg/kg. This led to collection of cloud droplets that was too rapid. In warm rain conditions this would create rain too quickly. The solution was to increase the number of lookup table value of “`nrrcr`” from 10 to 30 in `micphys.f90` and lower the minimum mixing ratio threshold to 1.e-12 kg/kg in subroutine “`make_autotab`.”

*Continued modernizing and streamlining the REVU package files, including fixing some of the LEAF patch variables. Substantially streamlined the dumpout or text output option. This will now be the “TEXT” option. Made all REVU files “implicit none” compatible and eliminated unused but declared variables. Fixed the section in “iplt.f90” that determines min and max dimensions based on variable type and output ranges in REVU_IN.

*Reorganized some REVU HDF5 output code to make things a bit cleaner looking, which should help with future updating. Made things such as subroutine calls and end and header conform to that syntax used in the rest of RAMS.

*Started organizing the code to place all “MAX” array limit variables in the same file named memory/grid_dims.f90. This way, future changes to max array sizes and be done in a single location for all purposes (ie. RAMS, REVU, ISAN, ODA, etc).

VERSION 6.1.07

*Made common variables “strl1” with character length 128 and “strl2” of 256 length that can be used throughout the model for declaring max string length.

*In isan_name.f90, rname1.f90, rname.f90, removed “maxrec” and modified call to “findgr”.

*Updated grid_dims.f90 as the central location for all “max” variables in the model code.

*In ref_sounding.f90, io_params.f90, var_tables.f90, mem_oda.f90, mem_mksfc.f90, mem_varinit.f90, isan_coms.f90, isan_name.f90, rname.f90, landuse_input.f90, removed “max” variables to be moved to grid_dims.f90.

*Few minor syntax changes to REVU.

*Streamlined and standardized how RAMS and REVU read in command line arguments and the runtime command execution.

*Standardized how “character” array lengths are declared. (There was a mix of old and new fortran syntax.)

*Changed several RAMSIN flag names to be shorter for code syntax purposes. These changes are made in all relevant files throughout the model:

1. wt_nudge_grid to wt_nudge_g
2. wt_nudgec_grid to wt_nudgec
3. roda_zfact to roda_zfac
4. oda_upaprefix to oda_upapref
5. oda_sfcprefix to oda_sfcpref
6. frqstate_keep to frqst_keep
7. itrackepsilon to itrkepsilon
8. itrackdust to itrkdust
9. itrackdustifn to itrkdustifn

VERSION 6.1.08

*Added a condition statement in sedimentation routine “sedim_trubin” to keep the array condition integer “idensrtgt” from going out of bounds. Should stay between 1-40.

*Added RAMSIN variables “IAEROLBC” and “AEROTAU”. Setting grid-dependent variable IAEROLBC to 1 will reset the initial aerosol vertical profile to the initial conditions along the lateral boundaries. This will function for zero gradient boundaries only (see LSFLG in RAMSIN namelist documentation). This would typically only be used on an outer domain, but could be used for nested grids if certain aerosol sources are needed. The grid dependent flag AEROTAU is a number in seconds that is used to compute the inverse time weighting. The time weight gets

computed as (timestep/AEROTAU) and lets the user control the strength of resetting the aerosol profile. If AEROTAU = timestep then full aerosol resetting occurs; if AEROTAU > timestep, the resetting will be more gradual.

VERSION 6.1.09

*Removed use of RAMSIN variable DELTAY since we should always have DX=DY in the model. Some routines (ie. mxdefm) actually only use dx or dy and not both in certain calculations. So a difference would produce inconsistent results. So, throughout the code I removed DELTAY and replaced it with DELTAX as needed. DELTAX is now simply the horizontal grid spacing.

*Removed use of NSTRATY which was the Y direction grid nesting ratio. Our nest ratios should always be the same in X and Y except for 2D simulation where Y nest ratios would always be 1. So I removed NSTRATY from RAMSIN and assign it in "opspec" where NSTRATY=NSTRATX except in the 2D case.

*Got rid of snow ingest variables "snow_water_sfc" and "snow_depth_sfc" and just ingest and use these products in the already declared variables "snow_mass" and "snow_depth".

*Fixed a bug related to ingest of snow depth variable. I had been assigning snow_depth equal to 5 x the "snow_mass" value but then unintentionally adding the ingested "snow_depth". We should be using one or the other and not both. Use 5x option when snow is not ingested and use the ingested "snow_depth" when available.

*Updated revu/hvlib.f90 and revu/textout.f90 to rename "snow_water_ps" to "snow_mass_ps". And updated revu/textout.f90 to apply correct variable string length for "snow_mass_ps" and renamed gempak abbreviation from SNOC to SNOM.

VERSION 6.1.10

*In applying history initialization, found that "leaf_class", "patch_area", and "soil_text" need boundary assignments made before the first analysis write and first timestep. So, these were added to mksfc_sfc.f90 subroutine "sfc_read".

*In ndvi_read.f90 subroutine "ndvi_update", removed the boundary assignment for veg_ndvic and veg_ndvip, since these are not necessary as "veg_ndvif" is the value stored in the NDVI surface files.

*In rbnd_nonscalar.f90 subroutine "leaf_bcond" removed setting boundary conditions for "patch_area", "leaf_class", "veg_ndvif", and "soil_text" since these are now set when the surface data files are read in during a model start.

*Removed use of TOPTA and TOPMA since these are redundant. TOPT = TOPTA and TOPM = TOPMA. Not sure why these were in the model since they were not used. These are also eliminated from model analysis files and REVU post-processing. Where TOPTA and TOPMA were used, they have been switched to TOPT and TOPM.

*In rdint.f90 subroutine "initlz" for history restarts, I removed calls to "grid_setup" and "make_sfcsfiles" since these are already called for any model start from subroutine "rams_master".

*In routine "inithis" began making incremental fixes and updates, but this is not yet functional.

*In revu code, changes "topo" to "topt" for consistency and removed "topta".

VERSION 6.1.11

- *In `rbnd_nonscalar.f90` subroutine “leaf_bcond”, added boundary setting for variable “veg_albedo”. This had been left out for setting bc’s in the J direction.
- *Added RAMSIN variable “NVEGPAT” to I/O for various nudging options and for history initialization and grid structure comparisons.
- *Commented out call to “prgintrp” for history initialization since we do not just want to interpolate from new parent grid to new nested grid. Rather we will interpolate from all relevant history grids to all new grids.
- *Made further changes to history initialization in subroutine “inithis”, including removal of unnecessary subroutines based on new method of interpolation to new grids. But this is still a work in progress.
- *Switched array dimensions for “soil_text” in surface files to be (i,j,k,patch) rather than (k,i,j,patch) in order to conform to the rest of the model output files.
- *Changed naming convention of “topo_z0” to “topzo” in topography files so that these conform to the rest of the model use of topographic roughness.
- *Changed naming convention of “veg_ndvi” to “veg_ndvif” in NDVI files so that these conform to the rest of the model use of NDVI.
- *Added “veg_albedo” in REVU output options. Removed “scalar1” and “scalar2” from REVU output options since these are obsolete.
- *Changed use of ISFCL conditions statements to only allow values of 0 and 1.
- *Changed NVGCON max to 20 in “rname.f90”. Changed vegetation class loops from (1,nvtyp) to (0,nvtyp) since types range from 0-20. Likewise, changes array dimensions of NVTYP variables to 0:NVTYP.
- *Removed top level initialization of “veg_rough” and “veg_albedo” in subroutine “sfcinit_nofile” since these values are overwritten in the following call to subroutine “vegndvi” that sets more realistic values.

VERSION 6.1.12 (HDF5 array swap in this version)

- *Cyclic BCs were fixed to produce duplication whether run sequentially or in parallel. This was done by Steve Herbener and required a large overhaul of numerous MPI routines calls to cyclic boundary conditions.
- *Updated the radiation BCs in “rbnd_nonscalar.f90” to include ‘rshort’, ‘rlong’, and ‘cosz’.
- *Included new integer variable “INITORIG” that tracks the original simulation type of INITIAL = 1 or 2 (ie. horizontal homogeneous or variable initialization). If we are running history initialization (INITIAL=3) we still need to know the original simulation type for many applications. The original value of INITIAL is stored in INITORIG and retained in output header files.
- *In `coriolis.f90` subroutines “corlsu” and “corlsv”, put in condition statement to exit the routine if running INITIAL=2 or INITIAL=3 and INITORIG=2. Needed for history initialization.
- *In `kf_rconv.f90` and `rconv.f90` I forced the KF or KUO convective parameterizations to only reset arrays to zero if time=0 and NOT a history initialization. Also set these parameterizations to run at times based on the same modulus calculation as we use to call radiation. Needed these to make history initialization match history restart when grid are the same.
- *In `rdint.f90` subroutine “initlz”, the variable INITORIG is zeroed and then set to the initial value to INITIAL if we are running INITIAL = 1 or 2. Also put in statement to only call BUBBLE if INITIAL=1.

- *In history_restart.f90 subroutine “history_start”, added i/o statement to include “INITORIG” in the output header files.
- *Made many modifications to inithis.f90 to get history initialization functioning.
- *In opspec.f90 subroutine “opspec3”, included statement to allow Rayleigh damping for INITIAL=3 if INITORIG=1 (ie. HH simulation).
- *In rcio.f90 subroutine “commio”, included INITORIG in i/o list of variables.
- *In rprnt.f90 subroutine “prtopt”, included statement to print initial sounding for INITIAL=3 if INITORIG=1 (ie. HH simulation).
- *In hdf5_utils.f90 subroutines “shdf5_info” and “shdf5_orec”, included the array switch for file output necessary for using Parallel HDF5, which is our future plan for RAMS in order to use the distributed memory version that Steve Herbener is developing. The array swap will allow direct access file opening by many software packages. The RAMS and REVU output files will now be in row-major format. “sdfopen” for Grads and “h5read” for Matlab can directly access the analysis files and REVU output files.
- *In interp_lib.f90 subroutines “gdtost2”, “htint”, and htint2” made a few modifications related to horizontal and vertical interpolation. These changes allow for more precise computations among code bases and machines that offer different precision depending on arithmetic order.
- *Minor change to output statements in grid_struct.f90 to be more readable.
- *Updated routine “check_real” to compare arrays based on old and current grids rather than just the old grids. Also commented out some sections of subroutines “cond_update” and “nud_update” so they will compile; these routines are not yet used.
- *Added variable INITORIG to memory in mem_grid.f90.
- *Changed “igrd_match” variable to be 2D to compare multiple grids from history and current domains where necessary for history initialization and nudging.
- *Added 3D precip rate arrays such as “pcpvr” to the negative adjustment routine so that these arrays are set to zero is associated hydrometeor mixing ratios are zero. Without this fix, it was possible to have precip rates non-zero where no hydrometeors exist, which could cause problems with aerosol precipitation scavenging.
- *In mpass_dtl.f90, added variable INITORIG to the put and get routines for parallel processing and updated the nwords count.
- *In rad_driv.f90 subroutines “radcomp” and “zen”, switched to a more precise and consistent computation of julian day that is needed for standard runs and history initializations runs to be consistent over time where the date changes in the simulations. Also included a more consistent day hour diagnostic that functions to find the time of day regardless of model start time. This was needed for history initialization to work precisely.
- *Fix to gaussian bubble implementation in subroutine “bubble” in ruser.f90 to resolve a bubble radius versus diameter discrepancy.
- *In turb_diff.f90 subroutine “truhor”, fixed a bounds check error that sometimes occurs related to calling and running subroutine “topobnd” for 2D simulations where topography does not really exist in the Y direction. The fix was to only allow the Y direction computation if simulations are 3D.
- *Minor change to the default idealized mountain topography routine in ruser.f90 where the user specifies a mountain shape for idealize mountain flow simulations. The default setup is more simple and universal.
- *Small print out changes to subroutine “textout” in textout.f90 to print null terminator strings on blank lines in gempak text format.

VERSION 6.1.13

Summary: This version largely improved the history file initialization option. It also removed the history file nudging, condensate file nudging, and recycle file initialization options in order to replaced these by creating history-varfiles that contain history nudging grids and condensate nudging grids. The “recycle” option for LEAF variables was made to work within the history initialization routines with the RAMSIN flag IPAST_LEAF, but it does not require a full history initialization.

*In `rams_master.f90`, included the option to create history-nudging files from history file data. The runtime for this is “MAKEHFILE”. It will make history nudging files for as many history files that are present in the specified directory under “VAR_HFILE”. Also removed calls to `nud_read` and `cond_read` since these subroutines are removed. All nudging (traditional varfile, history, and condensate) will be done via varfiles with a single set of varfile interpolation times.

*In `nud_analysis.f90`, fixed condensate nudging weights to be more appropriate. This should be experimented with for best future usage. Switched out `nud_type` so that this is either 0 or 1 (ie. no nudging or nudging varfiles).

*In `varf_update.f90`, added condensate nudging grids to be read in from varfiles. This is only possible from history-varfiles using MAKEHFILE to ingest history file data.

*In `rdint.f90`, modified code to fix nudging type options, history initialization and partial history initialization using the “recycle” option. The recycle subroutine was eliminated and the recycle flag “IPASTIN” was renamed to “IPAST_LEAF”. If IPAST_LEAF = 1, but history initialization is not turned on, then history initialization is called just for the LEAF variables. This can be done for the same grid configuration or a new configuration similar to the history initialization capabilities.

*In `history_start.f90`, added a check section and statement that warns the user if they are trying to restart a run at a time when all model grids are not present. This is not allowed.

*In `inithis.f90`, continued refining the code and added capability to work with the recycle feature that currently works for LEAF variables.

*Changed `io_params.f90` and `rname.f90` to reflect changes related to IPAST_LEAF recycle option and history and condensate nudging flags. We no longer need a RAMSIN flag to hold the name of history or condensate nudging files. This is built into the history varfiles.

*In `asgen.f90` added a variable to hold condensate nudging arrays to place in isan varfiles. Also created new routine “`nudh_driver`” to run instead of “`isan_driver`” in the event we want to create history varfiles rather than traditional varfiles based on pressure level data grids. The history varfile driver works similar to history initialization and can create history varfiles for a grid structure that differs but fits inside a history grid. History varfiles will contain the standard 5 upper level fields as well as condensate fields for nudging. They will not contain soil moisture, soil temperature, and snow fields that the standard varfiles contain since this is not compatible with soil initialization. If these are needed, the user would need to do history initialization or LEAF recycle.

*Eliminated the fortran files: `recycle.f90`, `nud_read.f90`, `cond_read.f90`, `nud_update.f90`, and `cond_update.f90` since these are obsolete with the new history varfile option. Also throughout the code I eliminated references to condensate and history file nudging times since these no longer exist.

*Moved subroutine “`check_real`” to `numutils.f90` and expanded it to do better checking between history grids and current grids for history initialization, history varfiles, and/or recycle features.

*Few other minor naming convention changes to make some code more distinct.

VERSION 6.1.14

*Applied lateral boundary conditions for precipitation rate in file bc/rbnd_nonscalar.f90 subroutine “micro_bcond”. Interpolation between grids can give non-zero precip rate values on grid boundaries that interfere with visualization and contouring.

*Added condition statement in opspect.f90 to produce error if user attempts to set RAMSIN variable NVEGPAT to something greater than NPATCH-1. Inappropriate settings will result in segmentation fault and potentially erroneous surface files and erroneous results from the LEAF surface model.

*Made modifications to file micro/mic_misc.f90 subroutine “adj1” that controls the microphysics negative adjustment or positive definite checking scheme. We had seen grid boundary points containing positive mixing ratio and zero number or positive number and zero mixing ratio for some hydrometeor types. This inconsistency was fixed. Further, if mixing ratio is adjusted in this scheme then number concentration is equivalently adjusted in call to “aelmic”.

*Added REVU variables for density potential temperature “theta_rho” and buoyancy relative to liquid condensate loading “buoyancy_liquid”. At this time, these should be used with caution and are experimental. This impacted files hvlib.f90, rcomp.f90, and textout.f90.

VERSION 6.1.15

*Added code related to the Loftus et al. (2008) convergence generation code for idealized initiation of convection. This code is in surface/ruser.f90 subroutine “conv_forcing” and it computes U and V tendencies that generate convergence. The code is called until either a certain time is met or a certain W threshold is met. To get the domain maximum W when in parallel, a computation of domain max “vertvel” was added to core/modsched.f90. Further, in the non-DM code I added calls to routines “master_getvertvel” and “master_putvertvel” in core/model.f90.

*Computation of W budgets in subroutine “boyanc” was reworked to prevent passing of unallocated arrays, which sometime caused segmentation faults. Also used allocation statements to limit stack memory requirement.

*In file core/radv.f90 subroutine “fluxlimits”, changed 3D locally allocated variables to allocatable variables, then used allocate/deallocate statements to keep extra memory off the stack. This should help limit occurrence of segmentation faults related to stack memory overflow.

*In core/rtimi.f90 subroutine “tend0”, added call to convergence routine “conv_forcing”.

*In cuparm files kf_cuparm.f90 and kf_driver.f90, I removed passing of micro 3D arrays. Rather these are now accessed via the “use micro” statement. If they are allocated, then they are copied to allocatable local arrays to be used. This also prevent segmentation fault related to passing unallocated variables. Made a few other minor changes to Kain-Fritsch code related to microphysics levels. We have microphysics LEVEL=0-3 in this version and not 4-6 which used to include the primitive dumpbucket scheme. Lastly, in cuparm/kf_rconv.f90 I added a few condition statements related to hydrometeor mixing ratios that may not be allocated.

*In init/rdint.f90 I added calls to “init_tracer” which is used for custom initialization of extra scalars or tracer variables. They can be on any grid and there can be any number of these. Subroutine “init_tracer” was added to micro/mic_init.f90.

*In io/anal_write.f90, got rid of use of pointers to prevent potential memory issues.

*In io/opspect.f90, added some condition statements for use of the convergence forcing code.

- *In io/rname.f90, added RAMSIN variables related to converging forcing and tracer code.
- *In isan/asgen.f90, eliminated use of “nullify” statements. Use deallocate instead.
- *In lib/rname.f90, changed the length of the read statement for RAMSIN input to be longer. This allows long path names in RAMSIN and prevents skipping long names at read time.
- *In isan/isan_coms.f90 and all memory fortran files, got rid of “pointer” statements, and replaced with “allocatable” statements. Also got rid of “nullify” and changed to “deallocate”.
- *Added memory/mem_tracer.f90. This replaces old use of mem_scalar.f90 which was not quite working properly before. Added allocation of tracer arrays in alloc.f90 and “use mem_tracer” where necessary. Updated calls to tendency allocation arrays to add in the tracer tendency arrays.
- *Eliminated calls to “c03” and “pc03” in mic_driv.f90 since these are generally unnecessary. Rather I directly call “enemb” to adjust number concentrations after cloud and ice nucleation. Also eliminated the associated subroutines from within mic_misc.f90.
- *In micro/mic_drive.f90, changed sedimentation tables from pointers to allocatables.
- *Fixed the fall speed coefficient for cloud and drizzle in mic_init.f90 from 3173 to 3173.e4. The new coefficient is appropriate for velocity units of m/s.
- *In mic_init.f90 I eliminated several other unused but computed variables and their associated declarations in micphys.f90.
- *In micphys.f90, declared RAMSIN variables related to convergence forcing code.
- *In mksfc/mksfc_driver.f90, added deallocation of “sfcfile_p” arrays. Perhaps this fixes the random segmentation fault that sometimes occurs when making surface files.
- *In mpi/mpass_dtl.f90 for the non-DM code, I added routines “master_getvertvel”, “master_putvertvel”, “node_getvertvel”, and “node_putvertvel” in order to pass around the domain max vertical velocity needed for the convergence forcing code. In the DM version of the model, I only had to add subroutine “set_vertvelmax_allnodes” to accomplish the same task.
- *In mpi/mpass_init.f90 I added variables related to convergence code and tracer code that needs to be passed to the nodes. Also updated “nwords” for buffering.
- *In mpi/node_mod.f90, changed pointer to allocatable.
- *In mpi/rnode.f90, changed associated variable to allocated variable. In the non-DM code, I added calls to “node_putvertvel” and “node_getvertvel”. In the DM code, I added call to “set_vertvelmax_allnodes” to pass the domain max vertical velocity information.
- *In rad_drive.f90, got rid of pointers related to cloud water or no cloud water being passed into subroutine “radcomp3”.
- *In surface/ruser.f90, modified Cosine-squared bubble routine to also allowed 2D bubble for 2D convective idealized simulations.
- *In turb/turb_ke.f90, changed associated arrays to allocated arrays.
- *In turb/turb_k.f90, eliminated some of the potentially troublesome pointers that are now declared as allocatable arrays in the memory routines.
- *In init/rinit.f90 subroutine “fldinit”, removed calls to “azero” since all of these fields have already been zeroed out at variable allocation time. Altered the associated calls to “fldinit” within file init/rdint.f90.

VERSION 6.1.16

- *Made a few cosmetic changes to the code in order to more easily track and cross-reference subroutine and function declarations with their associated calls.
- *Removed additional unused subroutines from the model.

*Altered some informational print statements so that duplicate print statements do not occur when running the model in parallel.

VERSION 6.1.17 – June 30, 2015

*Updated binned riming collection kernels to be directly compatible with newer set of fall speed power laws.

*Modified subroutine “col3” which treats collision-coalescence between rain and each ice species. Applying the Milbrandt-Yau (2005) approach, via Adrian Loftus’ adaptation to earlier versions of RAMS, to compute density of coalesced rain-ice so we can appropriately determine the destination category. Old RAMS version simply placed all mass and number into hail as the destination category.

*Updating of “col3” works well with the binned riming approach to ice collecting cloud droplets. The use of both options now helps retain mass and number in the graupel category since not everything is transferred to hail. Since this works well, I modified the destination category in binned riming subroutine “auto_accret_ice” to perform similar to the original riming approach in subroutine “col2”. These modifications appear to work quite well together and provide good partitioning among ice categories following ice-liquid collisions.

*Had to eliminate the micro budget term related to mass transferred to hail as the destination category for rain-ice collisions since the destination category can now be variable.

*Added tracer output options in REVU.

VERSION 6.1.18 – Sep 28, 2015

*From detailed testing of ATEX case (Saleeby et al. 2015) it was decided to reset cloud mass-diameter coefficient back to 3173.0 from 3173.e4, and reset cloud nucleation droplet size to the diagnosed value rather than the smallest allowed droplet. Also turned off the advection monotonic flux limiter until further testing can be done.

*Added a conditional if statement in mic_driv.f90 after the call to “icenuc” for the lines of code that determines the new depth, or K levels, of the pristine ice layer. This was causing segmentation faults when ice was turned off in the model.

*Eliminated boundary condition options IBND and JBND =2 and =3 since these do not actually exist in the model. IBND=JBND=4 for cyclic boundary conditions is now IBND=JBND=2.

*Updated the “reflectivity_all” option in REVU to include all hydrometeor species. Added an adaptation for ice species to assume spherical particles; this adaptation takes the non-spherical mass-diameter relationships and computes an “alpha” coefficient assuming spheres. This is done for each hydrometer type at each grid cell, and is diameter dependent. Eliminated the individual hydrometer reflectivity. These could be added back within the section for full reflectivity.

*In REVU, placed a lower limit on output data values. Anything with $-1.e-20 < X < 1.e-20$ is set to zero so that we do not have to output ultra-high-precision values. Some post software cannot handle these really small and high precision numbers that ultimately do not matter.

*In REVU, added a variable called “empty3d” in the event that post processing software requires place holder variables. This is necessary in Quickbeam radar simulator, in which a certain order of variables is required. If your simulations, for example, does not have drizzle turned on, then this variable is not available in REVU; but Quickbeam requires that the drizzle category not be empty. In this case we replace drizzle with “empty3d” which contains all zeros.

*Fixed declaration of NGBEGUN to integer in io/inithis.f90. This was causing model seg faults.

*Fix to file name output in REVU for the TEXT output option.

- *In hvlib.f90 switched to allocate/deallocate of 3D variables need for reflectivity calculation. Use of local 3D declarations overwhelmed the stack memory on some systems.
- *Updated/fixed surface wind convergence forcing code in ruser.f90 in order to allow user to input divergence rate in RAMSIN rather than the amplitude. (via Leah Grant and Sean Freeman)

VERSION 6.1.19 – Mar 14, 2016

- *Added the Simple Biosphere (SiB) model version 2.5/3.2 from Lixin Lu. This is turned on by setting RAMSIN flag ISFCL=2 and setting an initial CO2 profile in RAMSIN flag CO2_INIT. Adding SiB impacts memory routines, lateral boundaries, surface field initialization, nesting, etc. The only new 3D variable related to SiB is the CO2 concentration and associated tendency, “RCO2P” and “RCO2T”.
- *In aero_sources.f90, fixed dust model lofting median radius since it had been set to the mean mass radius instead.
- *Removed “patch_wetind” from the model since this LEAF-3 surface model variable was never used.
- *Removed “scrx” from mem_mksfc.f90 and mksfc_driver.f90 since this is never used.
- *Removed passing of unnecessary scratch arrays in nest_geosst.f90 and simply allocated and deallocated variables as needed.
- *In hvlib.f90, changed default minimum total mixing ratio from 1.e-4 to 5.e-4 kg/kg for both computations of cloud top temperature. This is rather arbitrary and it is best to get cloud top temperature from a satellite simulator.
- *In hvlib.f90, made a bug fix to the dust accumulation variable “ACCPDUST” and changed the multiplier from 1.e6 to 1.e5 to get the correct units.
- *In hvlib.f90, fixed output for all surface flux variables and eliminated unused or redundant output variables. The flux variables were multiplying by density, but the density multiplier is already accounted for within the model (ie. sflux_u, sflux_t, etc).
- *In textout.f90, renamed a number of output variable abbreviations to help distinguish between different units for the same variable to be output.
- *In leaf3.f90, eliminated passing of temporary variables (ups2, vps2, ths2, rvs2, pis2, dens2, zts2) into subroutine “leaf3” that can be allocated locally.
- *Modified leaf3_init.f90 to allow initialization of both LEAF3 and SiB. SiB uses the LEAF3 initialization by default, but then has additional initialization specific to its routines since SiB is much more sophisticated than LEAF3.
- *In leaf3_init.f90, eliminated passing of temporary variables (prsv, pis) which can be allocated locally and declared as individual reals rather than arrays. Similar change to initialization in ruser.f90.
- *Change in sfc_driver.f90 to merge subroutine “leaf3” and “sib_driver” into a single routine to prevent have a bunch of redundant code.
- *In leaf3_init.f90 subroutine “sfcinit_nofile” and ruser.f90 subroutine “sfcinit_nofile_user” and inithis.f90 subroutine “sfcinit_hstart”, added code to prevent soil_water (ie. soil moisture) from exceeding 100%. This can occur due to difference between land surface soil types in RAMS and the data source model and when history initializing on a different grid. Also adapted this difference between LEAF3 soil type porosity and that used in SiB.
- *In SiB and RAMS micro, modified some code that was used to prevent division by zero. In some places, denominators were added by small numbers such as 1.e-20 or 1.e-12. However,

some compilers will take $(0.0 + 1.e-20) = 0.0$, which will give a NaN when used as the denominator. Rather, we will be using the “max” function everywhere we need this, such as “max(variable,1.e-12)”. This has not been altered everywhere in the code, so when NaNs show up, be on the lookout for this type of error.

*Fix to array bounds potential error in subroutine “htint” related to array “eleva”. The array bounds error does not impact model solution since it occurs only when the associated calculation has zero in the denominator. The array bounds error value generated a NaN that would cause the model to crash.

CONTINUE BELOW FOR UPDATES STARTING WITH RAMS VERSION 6.1.20

MODEL UPDATES STARTING WITH VERSION 6.1.20 WITH ADDITION OF HUCM-SBM BIN MICROPHYSICS AND SIB CARBON-CYCLE LAND SURFACE MODEL

VERSION 6.1.20 – July 13, 2016

*Added HUCM Spectral-Bin Microphysics Model. (via Adele Igel). This impacts several parts of the code. Added subroutine “wetthrm3_bin” in rthrm.f90, condition statements in rtimh.f90 for bin-micro, condition statements in rdint.f90 for bin-micro, new output variables in anal_extra.f90, extra output conditions in anal_write.f90 for bin-micro, extra conditions for history starts in history_start.f90, condition statements for flag checking in opspect.f90, hdf5 conditions in isan_io.f90, conditions for hdf5 output for 4D bin micro in hdf5_utils.f90, changes to memory allocation for 4D bin micro with additional dimension of number of bins, inclusion of bin-micro memory allocation in mem_micro.f90, increased size of scratch arrays in mem_scratch.f90 to accommodate 4D bin-micro, adding tendency arrays for bin-micro in mem_tend.f90, condition statements in mic_misc.f90, mpi consideration for buffer sizes of 4D variables in bin-micro within mpass_cyclic.f90, mpass_init.f90, mpass_lbc.f90, node_mod.f90, and para_init.f90. Modification to rad_driv.f90 to allow bin-micro to interact with Harrington radiation. Additions to turb_k.f90 in calls to “ae1” and “sum_bins”.

*Bug fix to “Gaussian” warm bubble (IBUBBLE=2 option) temperature/moisture perturbation in ruser.f90 subroutine “bubble”.

*Fix to random temperature perturbations for IBUBBLE=3 option. The random bubbles were not initialized correctly in the DM code since each sub-domain was running the random number

generator. Now, only the main node will compute the random number and then broadcast them to the other nodes. Also changed the default random bubble generation to be a 0.1K maximum temperature perturbation that is maximized at the surface and decreased linearly to 500m altitude. (via Leah Grant and Steve Herbener)

- *Added code to allow SiB to compute the surface albedo and upward longwave radiation rather than using the LEAF3 default code when SiB is run. This impacted `sfc_driver.f90`, `sib2_co2.f90`, and `rad_driv.f90`. Both the `sfc_driver` and `rad_driv` call the subroutine “`sfcrad`”, which should not be called if running SiB.

- *Turn advection monotonic flux limiter back on by default. This is only for the vertical direction.

- *Added base state density (DN0) to the extra output variables in analysis files.

- *Fixed some errors in the budget variables related to latent heating by condensation and freezing. Also fixed the total melting budget variable to prevent double counting of some melting. Latent heating budgets are set to report $d\Theta(K)$ by default. This can be changed in `micphys.f90` variable “`lhrtheta`”. If set to “false” the latent heating budget variables will return $dTemperature(K)$. (via Adele Igel)

- *Added a lot more comments to SiB and reordered some SiB variables to group them by common function. Also linked SiB data to the LEAF3 output variables `VEG_TAI`, `VEG_LAI`, `VEG_ROUGH`, `VEG_HEIGHT`, `VEG_ALBEDO`, `VEG_FRACAREA`. And computed `PATCH_ROUGH` via SiB info rather than using LEAF3 defaults. These fields are 0 for first analysis file since SiB is not called until first timestep.

- *Added call to “`non_scalar_bc`” before writing to LITE, MEAN, and BOTH file types.

- *Prevent automatically writing the “extra” variables in `anal_extra.f90` to LITE, MEAN, and BOTH files unless clearly directed to do so in RAMSIN. Did this since at LITE writing time, only LITE variables requested will be returned to master in non-DM code unless it's a full analysis write time. See notes in `anal_write.f90` to override this.

- *Added accumulation and accumulation rate for total aerosol mass. This had already been done for accumulation of dust-only. Note that this accumulation is from rain-out only and not dry deposition of aerosols to the surface. That is still work to be done, but the dry deposition rate is quite small compared to rain-out for short term simulations.

- *Updated history initialization of surface quantities for SiB modifications. Prevent interpolated soil moisture from exceeding 100% due to altered soil porosity via interpolation to new grid. Avoid interpolating land surface class information since this can produce non-integer values when integers are expected. Reset a number of variables (ie. roughness, stomatal resistance) since leaf class may change via new-grid interpolation.

- *Added recycle feature to SiB land surface variables.

- *Altered hard-coded path to supplemental files such as `DustEmission.dat`, SiB land info dataset, and HUCM input files. Set this to “`../etc`”.

- *Added HUCM model and SiB model REVU output variables and REVU net radiative flux computation. Fixed units for REVU variable `ACCPDUST` (surface accumulated of dust through wet deposition; not thru dry deposition).

- *Modified subroutine “`vegndvi`” to create separate routines “`veg`” and “`ndvi`” in order to have these work well with both LEAF and SiB land surface models.

- *Updated HUCM bin-microphysics model to work for history restarts and history initialization and history varfiles. This affected `rdint.f90`, `inithis.f90`, and `asgen.f90`.

VERSION 6.1.21 – August 25, 2016

- *Removed SiB diagnostic variable NEE (net ecosystem exchanged) from RAMS and REVU since this is essentially a duplicate of CO2FLX (flux of CO2 from canopy). Removed from sib2_co2.f90, sfc_driver.f90, nest_geosst.f90, rbnd_nonscalar.f90, hvlib.f90 and textout.f90.
- *Added code to initialize CAS CO2 (pco2ap) SiB input variable based on initial CO2 (co2_init) and air pressure rather than hard coded value in subroutine “init_sib”.
- *Split SiB CO2 and PCO2AP initialization into two subroutines so that I can only initialize PCO2AP when doing history restarts with added grid. The 3D CO2 variable is automatically interpolated on history restart with added grid, so we do not want to reinitialize CO2. The reinitialized PCO2AP is overwritten if RAMSIN flag NOFILFLG=0, but is needed if NOFILFLG=2.
- *Eliminated use of NOFILFLG since only option = 2 really worked. This was used in subroutine “geonest_nofile”. It was called at model initialization and for history restarts with added grids. This subroutine is now only called at model initialization and not for history restarts or history initialization. For history initialization or history restart added grids, the “geonest_nofile” variables are interpolated from previous or parent grids using tested routines in inithis.f90. If the user needs to truly reset the surface data via “geonest_nofile”, a call to “geonest_nofile” or “sfcinit_nofile_user” can be added to initialization.
- *Eliminated “HISTORY” restart section in rdint.f90 subroutine “initlz” since there was much redundant code and since the old interpolation of history added grids does not seem to match the correct interpolation used in history initialization. The history restart was integrated into the history initialization code in inithis.f90 when grids are added at history restart; otherwise “history_start” is still called. Modified inithis.f90 to incorporate history restart added grids by preserving the original start and restart times and model duration and dealing with variables that do/do not need to be reset on history restart. Code was added to rams_master.f90 in non-DM version and rams_model.f90 in DM version to determine if doing history restart or history initialization. History header is read in here rather than later on as previously done. A history restart causes flag INITIAL to be set to 3 if grid is added. Flag checks in opspec.f90 have been adjusted accordingly for soil initialization using ISOILDAT=1.
- *For initializing with ISOILDAT=1 in the DM code, we force the model to use a single node for the soil moisture/temperature and snow field assignment. This has to be done on 1 node due to nearest neighbor interpolation. This is done in rams_model.f90. This cannot currently be done correctly on multiple nodes. The user has to run the initialization and the model will output the first analysis file. Then a parallel history restart can be run on many nodes.
- *Made a bug fix to history initialization interpolation of WP and WC. Had been interpolating via T-level information, but needed to be using M-level information for vertical interpolation. Horizontal interpolation is fine on the T-grid and M-grid for scalars and momentum variables. This required modifying input variables in all calls to “hi_interp” throughout the model.
- *Eliminated code for “patch_land_average”, “patch_land_unaverage”, and “patch_interp” since these are not being used for interpolation. This could be incorporated into history initialization and history restart added grids for a smoother interpolation of land surface characteristics between grids, but it does not appears to offer substantial gain. This could be examined more closely in the future.
- *Eliminated code in mksfc_driver.f90 related to surface file checking during history restart. This is redundant code and is now done at initialization for all grids. Also made a bug fix in use of the RAMSIN namelist flag “isstflg” in the section on NDVI; changed this to used the flag “ndviflg”.

- *Add SiB CO2 lateral boundary forcing for zero gradient LBCs to prevent CO2 from depleting too much over time. This simple forcing just forces LBCs to past value using a weighting function that is applied to one interior boundary perimeter. This is ok but not smooth. This is rather brute force and should be done more like varfile nudging in the future.
- *Changed call to “stars” to only be called (when running SiB) if patch=1 or (initial=2 and time<0.001) or patch_area<0.009. Had to move computation of “ubmin” and “vels_pat” out of subroutine “stars” since the routine “sfclmfv” always needs these variables. Making this change speeds up the model by not running “stars” un-necessarily for SiB.
- *Altered naming convention for recycle variables from “recycle_leaf” to “recycle_sfc” and RAMSIN variable “IPAST_LEAF” to “IPAST_SFC” since these are no longer exclusive to LEAF, but can also be used with SiB.
- *Made a bug fix for calls to “grndvap”. Calls to this routine should be passing the top snow layer or surface water layer data to the routine (layer ksn). Rather, it had been passing in layer (mzs) which is the maximum number of snow or surface water layers. However, KSN does not always equal MZS. This could cause serious errors in fluxes from snow surfaces when MZS in the RAMSIN namelist is > 1.
- *Fixed REVU variables related to surface snow depth, mass, and temperature. Had to fix sections of hvlib.f90 and rcomp.f90 to make the output correct. The previous subroutines were wrong in their variable passing and computations of these variables. I do not think that they ever previously worked correctly.
- *Got rid of calls to “snow_init” and deleted this subroutine. It is not necessary since idealized snow initialization can easily be added to ruser.f90.

VERSION 6.1.22 – September 12, 2016

- *Altered use of NGBEGUN in history restart/initialization. For history initialization or history restart added grid, NGBEGUN will equal 1 rather than 0 if they are being restarted from a previous simulations that was beyond time=0. Since these grids are interpolated from other grids, they do not start from a time=0 sort of state, meaning that past and current momentum and pressure fields are different. From a true time=0 state, the past and current are equal and hence RAMS uses a different Asselin “filter parameter” for the first timestep in order to preserve amplitude in this special case.
- *Fixed a bug in history restart added grids for 2D fields.
- *Added namelist parameter ISNOWDAT which is similar in function to ISOILDAT, but gives the user control over whether to initialize snow fields from data in varfiles.
- *Eliminated “alloc_leafcol” in which some 1D variables are dynamically allocated for LEAF. Rather than have a separate awkward module/routine for allocating these, I added these variables to the “leaf_coms” variable declaration and gave them static array sizes of “nzgmax+nzsmax+1”. These are not large arrays, so this should be no problem.
- *Got rid grid dependent polelat and polelon (ie. platn(ngrids), plonn(ngrids). This was an unnecessary lingering grid dependent variable that had hung around since the days in which RAMS was trying to be turned into a dual-hemispheric (global) model. This never worked correctly and was replaced by OLAM. So, I decided to remove the grid-dependent nature of platn and plonn and just use polelat, polelon from the RAMSIN namelist.

*Fixed radiative heating rate in Harrington and Chen-Cotton schemes so that the heating rates are in potential temperature rather than temperature. This simply required dividing the heating rate by the exner function. Set top and bottom boundaries for radiative heating rate “FTHRD”.

*Moved the setting of the top and bottom boundaries for SWUP, SWDN, LWDN, LWUP to “radcalc3”. These are computed on “m” levels, so it was best to set these inside radiation on radiation levels. The top boundaries are set to the top radiation level which may be further aloft than model top. This is useful for computing radiative balance.

*Fixed code that was commented out to allow analysis write of “extra” variables to LITE files that are the LEAF type 4,5,6 that have dimensions that include patches, soil layers, or snow layers. This had not been functioning, but does now. A couple of non-active placeholder LEAF variables are in the code in “anal_extra.f90” and “anal_write.f90”. This has to be customized as needed and care should be taken when adding “extra” variables since the numbering process must be exact.

*Fixed an error in the Chen-Cotton radiation subroutine “stable” that some compilers do not like. The issue was a DO loop that did not loop. The loop did something like DO N=K1,K2 where K1=K2. One compiler did not like that and decided not to run the code in the loop, which led to a division by zero and Nans and a model crash. I eliminated the loop and just ran the code that was within the loop.

VERSION 6.1.23 – November 11, 2016

*Added “rvt0” as an additional 3D base state moisture that is linked to “rt01dn” 1D base state variable. Needed this in order to experiment with diffusing perturbations of RTP in a non-homogeneous environment with topography. (only in non-DM codebase)

*Added bug fix to vertical interpolation routine “htint”. This routine would allow a new model grid to extend higher than an old grid, thus requiring extrapolation from the old grid top to new grid levels. This extrapolation could allow for negative moisture. The bug fix was to simply prevent new grids from exceeding the model top of older history grids.

*Separated out subroutine “ugetarg” from rsys.F90 and created new file rget.F90. This new file with “ugetarg” is also used by newly implemented data preparation code “dprep” that is added in this version.

*Added degribbing code to RAMS in the “dprep” model directory. There are versions to decode Grib-1 and Grib-2 formats which are very different. Both require compiling 3rd party degribbing software in order to call the executables “wgrib” or “wgrib2”. The degribbers are compiled in “bin.dataprep1” and “bin.dataprep2”. These codes take grib formatted gridded datasets such as GFS, NARR, ECMWF, RAP, etc. and create the RAMS “dp” files in RALPH-2 format (see documentation). These formatted datafiles are used in RAMS to create grid-dependent variable initialization files (or varfiles). These programs named something similar to “dgrib1_main.f90” and “griber_grb1.c” have to be customized for new input grib datasets since all grib datasets vary by grid type, data naming, and units.

*Changed the text output name abbreviation of total mixing ratio variables from REVU. The old names conflict with GEMPAK function names, thus leading to errors.

MODEL UPDATES STARTING WITH VERSION 6.2 WITH ADDITION OF FULLY PARALLELIZED FEATURES THAT ALLOW 1-WAY GRID NESTING IN THE DISTRIBUTED MEMORY (DM) VERSION OF RAMS. FROM THIS POINT FORWARD WE WILL ONLY SUPPORT THE “DM” VERSION AND WILL

DISCONTINUE UPDATES TO THE TRADITIONAL CODE VERSION THAT REQUIRED A MASTER NODE AND FULL LOADING OF 3D GRIDS INTO MASTER MEMORY.

VERSION 6.2.01 – December 12, 2016

*Added Steve Herbener's code updates to allow the DM code to include fully parallelized 1-way grid nesting that is needed for current supercomputing systems with limited amounts of memory per node. Stopped updates to the standard non-DM version of RAMS.

*Removed code related to 2-way grid nesting (nested feedback) for simulation runtime. Two-way nesting remains for surface and varfiles to deal with topography differences between nested grids. The implementation of 2-way nesting was non-functional in the DM and would require a large development effort to run in parallel on massive supercomputers.

*Minor fixes to density weighting of 4D var and isan variables as well as a fix to grid nesting of the vertical velocity field.

*Added additional lateral boundary settings necessary for history initialization on different grids.

*In the DM code, moved soil moisture quality control checking to the ISAN stage since this requires knowledge of the full domain in order to run. Since the making of surface and varfiles is forced to run on a single processor, this seemed the most efficient place to run soil moisture checking.

VERSION 6.2.02 – February 10, 2017

*Added feature to allow REVU to post-process surface model variables into HDF5 post-processed output. A values of IZTRAN=4 has been added to REVUIN to indicate processing variables on soil levels / snow levels / patches. IZTRAN=4 will only work with surface model variables in which ivar_type = 4, 5, 6 in hvlib.f90. IZTRAN = 1,2,3 will only work with ivar_type=2,3 in hvlib.f90.

* Fixed bug in production of some "extra" variables in LITE files. There was a potential redundancy in writing DN0 from "extra" variables. This cannot be allowed for variables that are already allocated in the variable tables list.

*Eliminated the variable timestep option "IDELTAT" since this never functioned very well. User must choose a timestep.

*Updated dataprep code to ingest additional gridded reanalysis and forecast grids with specific variable names. This is an ongoing process in the de-gribbing code as new dataset options are added. As usual, this is customized, so the user must know precisely what data is coming from gridded reanalysis, what projection the data is on, and what the variable names are, as well as whether the format is grib1 or grib2. Updated griber_grb1.c, griber_grb2.c, dgrib1_main.f90 and dgrib2_main.f90.

*Added nudging of soil moisture and soil temperature in order to help control the soil state for long term simulations. This might be needed if the land surface models dry out the soil too much over time. We only have limited soil depth and layers, so this type of internal soil nudging could be useful in some regional climate simulations. Added SNUDCENT to RAMSIN namelist for soil moisture nudging time scale.

*Added the source code for making RAMS surface data "block files" to the main source code directories and streamlined this code from previous versions.

*Eliminated the need for utils_sub_names_nondm.h include file and combined RAMS hdf5 utilities files as I was able. RAMS and REVU now access a number of common hdf5 utilities

routines. This could allow REVU to be parallelized in the future. For now REVU is hard coded to access the full domains via special “ramsorrevu” flagged section in hdf5_utils.f90.

- *Added flag IIFN_FORMULA so the user has control over what formula is used for heterogeneous ice nucleation. There are the general DeMott et al. (2010) and the dust formula from DeMott et al. (2014).

- *Updated the variable MAXMACH in grid_dims.f90 to 2048. This file contains a subroutine to set some max values for allocation of some variables that cannot easily be allocated dynamically.

- *Fixed a non-looping DO loop in Chen-Cotton radiation that was failing and crashing the model with some older PGI compilers.

- *Removed the options for IDIFFK=5,6 since the “experimental” implementations have been tested and appears to be non-functional. These could be revisited at a later time.

VERSION 6.2.03 – February 10, 2017

- *Reduced memory allocation for MAKESFC and MAKEVFILE to prevent allocation of full model scalars in alloc.f90. A separate routine was added to only allocate the scalar arrays and modules needed for making surface and varfiles. This impacted nesting interpolation routines and required creating separate ones for these runtypes.

- *Modified code to allow RAMS, REVU, and Degribbers to work with PGF90, IFORT, and GFORTRAN compilers. This involved creating several separate subroutines in REVU for HDF5 outputting with GFORTRAN since GFORTRAN does not seem to like the “optional” declaration statement in subroutine.

- *Fixed a few minor bugs and bad coding practices that came to light during recent testing.

- *Modified Makefiles to join together the object make file and standard Makefile. Then changed the Makefile to remove suffix rules and add pattern substitutions with the help of Steve Herbener. This allows us to compile each fortran file with a custom optimization as needed. On my latest test machine and compilers, I could only get run-to-run duplication with optimization –O2 if I set the optimization of kf_cuparm.f90 separately to –O1. The reason for this is unclear, but this offers a solution for now.

- *Minor technical changes to mk_main.f90 for block file creation.

- *Small fixes to KF cuparm scheme to prevent mixing of integers and reals.

- *Small changes to de-gribbing code to fix reading in of integer values.

- *Some changes to rcio.f90 for reading the RAMSIN namelist single integer and float arrays.

- *Small changes to memory allocation in isan/asgen.f90 to add consistency in sub-domains.

- *Bug fix in vertical interpolation routine in lib/interp_lib.f90 that can sometimes impact domain top if extrapolation is used rather than interpolation.

- *Added LEAF variables allocation to memory/alloc.f90 in order to keep all allocation in the same place.

- *Fix to history restarts using HUCM-SBM microphysics in micro_bin/microphysics.f90

- *Bug fix in mksfc/ndvi_read.f90 related to reading the NDVI surface files in parallel.

- *Added nesting routine “interp_fine_grid_sfcvar” to nesting/nest_drivers.f90 to keep surface nesting separated out from other nesting routines. This helps in dealing with LEAF and SiB for nesting purposes.

- *Added nesting routine “interp_varsfc” to nesting/nest_intrp.f90 to deal with nesting of surface variables when making surface and varfiles.

- *Streamlined some of the “rhdf5” calls and routines in revu/RAMS_hdf5.f90 and revu/rhdf5_utils.f90 to simplify this code used to write REVU output in HDF5 format.

*Minor array changes to SiB that some compilers are picky about.

VERSION 6.2.04 – April 11, 2017

*Changed analysis write formatting statement to prevent bug from occurring in which the number of grid cells is not written correctly to header file because the number is too large. This becomes a problem as we move to much larger simulations.

*Fix to call to “grndvap” related to the variable “ksn” and “nzs” which is the number of surface water/snow layers. Fix use of snow levels in leaf3_soilveg.f90, leaf3_init.f90, ruser.f90, inithis.f90.

*Fix to aerosol deposition in micro/aero_deposit.f90 related to potential division by zero error if relative humidity > 100%.

*Prevent dust lofting in micro/aero_sources.f90 for grid cells within 5 points of the boundary or within the lateral boundary nudging zone. We had found that dust lofting in these regions could produce excessive runaway dust production.

*Bug fix to the solubility of aerosols calculation for very small soluble fractions.

*Fix to potential array allocation error in radiate/rad_mclat.f90.

*Eliminate hardcoded value of “ubmin” in sib/sib_co2.f90. Rather, use the value from the surface driver from Leaf-3.

VERSION 6.2.05 – June 28, 2017

*Added new warning message to encourage users to set GNU = 4 for cloud and drizzle. For less narrow shape parameters for these 2, there can be a problem with gamma distribution truncation that prevents the correct amount of water being transferred to rain via collection.

*In mic_init.f90 and mic_tabs.f90, changed some of the default lower and upper size bounds on many of the hydrometeor types. Moved rain dmb0 from 0.1mm to 0.3mm. This prevents some truncation error for cloud and drizzle and rain and helps prevent some hydrometeors from becoming too small, which tends to allow some cloud to unrealistically persist over time due to microphysics/dynamics interactions.

*Bug fix to mic_vap.f90 to have some calculations go all the way down to k=1 level. This helps prevent some potential division by zero errors.

*Bug fix to array specifications in rad_aero.f90 and rad_driv.f90 related to looping through arrays that have not been set over the full array length.

VERSION 6.2.06 – September 13, 2017

*Fix to boundary conditions for some radiation parameters. Does not change model solution. Just prevents having a bunch of zeros along the boundary which is annoying when making full domain plots.

*Added a feature that allows users to choose to do normal scalar diffusion or do diffusion of perturbations from the base state, or diffusion of perturbations from the domain mean state. The domain mean state is computed each timestep. The diffusion of perturbations, when activated, only does this special diffusion for THP (theta-il) and RTP (total water). Added routines and code to utils_sub_names.h, parlib.c, mem_basic.f90, mpass_init.f90, rnode.f90, turb_diff.f90, and turb_k.f90. There is no RAMSIN flag to turn this on. This is for specialized research purposes and can be turned on by setting the flag IDIFFPERTS in the file micphys.f90. For the future, we should also add diffusion of perturbations from the base state winds. For now we assume the base state winds are zero as is commonly done in RCE simulations.

- *Fixed a potential bug in `rdint.f90` related to initializing dust source code without having done memory allocation. Otherwise the model would simply stop. Added conditions statements in `mem_micro.f90` to help deal with this allocation/usage issue of some dust tracking flags.
- *Added condition to `inithis.f90` in which history initializations require that the new grid use the same POLELAT and POLELON as the original grid.
- *Added numerous fatal error and warning statements in `opspec.f90` to help prevent users from mistakenly choose bad RAMSIN flag combinations. However, this only prevents some of the worst violations. Users still need to be aware of their settings and the consequences of their settings of RAMSIN flags.
- *Added RAMSIN flags that allow users to set up and run Radiative-Convective-Equilibrium (RCE) type simulations without having to make hardcoded changes. These are IRCE, RCE_SZEN, RCE_SOLC, RCE_UBMN, RCE_BUBL. This seemed like a good idea given the number of users running RCE simulations. RCE setup impacts `rad_driv.f90`, `rrad2.f90`, `ruser.f90`, `sfc_driver.f90`,
- *Added RAMSIN flags related to specifying the paths to files within the “etc” directory. These are DUSTFILE, SIBFILE, and HUCMFILE. This prevents having to hardcode the location of these required input files.
- *Added RAMSIN flag IDUSTLOFT to allow user to choose which type of dust lofting method to invoke. Added output of variable DUSTFRAC which is the fraction of erodible surface across a domain grid cells. Can be from 0 to 1.
- *Modified the dust lofting routine in `aero_sources.f90` to streamline the lofting via the Ginoux global dust source database.

VERSION 6.2.07 – November 6, 2017

- *Added diffusion of perturbations from varfile state. In subroutine “nudge”, computed varfile tendency of THP and RTP in `th00` and `rvt00` variables. Added IDIFFPERTS flag to RAMSIN for diffusion of perturbations. Make IDIFFPERTS=3 for diffusion from varfile state perturbations.
- *Updated dust data ingest in `data_init.f90` to be able to ingest large specific dust source locations and map them to RAMS grid without having to necessarily pre-define the grid size and statically compile the dust file array sizes. See the subroutine `handle_dustsource` for details.
- *Change IDUSTLOFT limit value to something larger than 1.
- *Eliminate static declaration of dust source grid array sizes in `aero_include.f90`. This is now done in `data_init.f90` to correspond to IDUSTLOFT flag in RAMSIN.
- *Update dust source options in `aero_sources.f90`.
- *Pass variable dust source data base array sizes into subroutine `broadcast_dustsource` for MPI broadcasting to compute nodes.
- *In `rad_drive.f90` treating stop conditions differently. Do not stop for “temperature too low” condition, but rather print warning message to standard output. Many times this error only occurs for a couple of time steps and then ceases to be a problem. The choice of the stop value is rather arbitrary anyway and was only input in the radiation when Jerry Harrington designed that code and was testing for errors.
- *Set upward longwave flux at the $k=1$ level (radiation surface) to “rlongup” from surface model. This is an important condition since the radiation model can produce different values from the surface model since they use two different methods to obtain the surface value of upward longwave flux at the surface. Radiation fluxes are on “M” levels, so $k=1$ is the surface.

*In `rrad2.f90` subroutine `flxlw`, examined the “negative source” error message. This seems to occur occasionally in the 8th radiation band at the uppermost model levels when the “`sigu`” and “`sigd`” values become slightly negative. Values overall tend toward zero at the upper levels sometimes; I believe precision in the polynomial curve fit cause the small negative values to sometimes occur. For now, I have turned off the model stop condition but still print the warning message to the standard output. In most cases the model gives the error message for a couple of timesteps and then proceeds forward without problems.

VERSION 6.2.08 – May 9, 2018

*Add code from RAMS 4.3 that computes the full Exner Function based on Medvigy et al. paper. Initial implementation appears to work but produces spurious results. This needs more examination and testing before use, and should be considered experimental at this time. The file is named `rexev.f90` and the calls are done to “subroutine `exevolve`” in `rtimh.f90`. Calls are currently commented out.

*Added an absolute value function for a “`dpdt`” statement in `kf_driver.f90` in the event of very small negative pressure change. This was already done in several places in the Kain-Fritsch code and a recent test triggered an error with this new computation.

*Changed the “temperature too low” error message in the Chen-Cotton/Mahrer-Pielke radiation routines to match that in the Harrington radiation. Model does not stop, but rather produces warning message.

*In the past, surface fluxes from the land surface model (ie. `sflux_r`, `sflux_t`, `sflux_w`, `sflux_u`, and `sflux_v`) were applied to the lowest model atmospheric level through the turbulent diffusion schemes. It seems that this could create a disconnect between what the surface model says the surface fluxes should be and what the atmospheric model actually sees. So, a change was made to add the surface flux tendencies to the END of the diffusion code in `turb_diff.f90` subroutine “`diffsclr`” for THP and RTP rather than earlier in the code via the surface flux variable “`sfcflx`”.

*Modification to tru-bin sedimentation so that the IDENSRTGT index is not determined by nearest-integer “`nint`” method, but rather by interpolation between index values. This creates a smoother vertical profile of falling hydrometeors and eliminates potentially spurious hydrometeor layering when using fine vertical grid spacing.

*Eliminated obsolete “`mpt2`” and “`mpt3`” flags that used to be used by `VTABLE`, but not really needed anymore.

VERSION 6.2.09 – Oct 12, 2018

*Primary change in this version is the addition of the KPP mixed layer ocean model. This is currently experimental and requires some customization, but results look promising. We still need to add profiles of ocean temperature, salinity, and U,V ocean currents as ocean “`varfiles`”. KPP works for history restart, but needs to be added for “history initialization”. Customization is required in `kpp_init.f90` for initialization of these 3D ocean fields. Call KPP timestep from `rtimh.f90`. Call KPP initialization from `rdint.f90`. Had to add ocean depth variables on ocean grid levels and declare them as “`idim_type=10`”. Add several KPP control flags to the RAMSIN. Had to add KPP levels to analysis file header similar to how we do this for atmospheric grid levels. Had to add new “`case`” statements for HDF5 utilities for this new “`idim_type=10`” for ocean level data to be written. New memory allocation section based on ocean levels and KPP datatypes. Added new MPI routine to compute and output the KPP maximum ocean mixed layer depth to standard out as a useful diagnostic. New KPP flags and ocean level variables with

“idim_type=10” added to MPI routines. Added surface model water internal energy condition statement depending on if KPP is being run.

*Multiple updates to REVU including many new parameters to compute for output. Also updated output to pressure levels potentially every 25mb from 1050mb to 50mb.

*Added variable flag and condition statement for potential interpolation in REVU to pressure levels below ground. This was done for offline aerosol AOD computation data, that was formatted to be able to run with REVU, since some extrapolation led to non-realistic negative values of AOD.

*Updated RAMS surface data block file computation to be able to process a specific ocean depth bathymetry dataset for the KPP ocean model.

*Included in REVU the potential to output AOD variables that are computed from RAMS aerosol data and run thru some Python-based AOD computational routines that output various AOD and aerosol extinction variables. If the Python code is run for ALL simulation levels, then the output can be formatted to run happily with REVU. The Python code was largely adapted to work with RAMS by Sam Atwood in Prof. Sonia Kreidenweis’s group at CSU.

*Fixed computation bug with latent heat of freezing budget variable “latheatfrz” related to melting. Parentheses were missing when adding in the partial melting of mixed phase species.

VERSION 6.2.10 – Mar 15, 2018

*Updates to degribbing code “dgrib1_main.f90” and “dgrib2_main.f90” and “griber_grb1.c” to handle new ERA5 data and to fix ERA-interim and GDAS data ingest related to different variable names and snow units specifications. Also added capability to ingest ERA5 on a limited area domain in addition to global data. Modified RAMS ingest of such data in “astp.f90” that is on a limited area domain but still uses a lat/lon grid (“inproj=1”); this is a special case.

*Modified fall speed coefficients for cloud and/or drizzle droplets which is used in hydrometeor vapor diffusion and evaporation routines. Still do NOT let cloud droplets sediment/fall as they are assumed too small to appreciably fall (per Walko et al. 1995).

*In subroutine make_autotab, re-adjusted smallest rain/ice mixing ratio bin back to $r3min=.01e-6$ and $rimin=.01e-6$. The former smaller values were producing inconsistencies in some of the lookup tables that were highly non-monotonic when they should be monotonic.

*Via testing by Adele Igel, we modified the method of determining how much ice species mass is involved in the riming process. New method might be a modest overestimate, but former method was apparently an underestimate. The amount of rimer mass involved in riming collisions with droplets helps determine the destination category of the combined collisional mass. Using greater ice mass as the rimer tends to shift rimed mass more to the snow/aggregate categories and less to the graupel/hail categories. Tests suggest that this is a positive change to the model. This introduced new subroutine to the method of moments called “sxyice”.

*Fixes to REVU when outputting variables on pressure levels. There was a memory allocation problem associated with situations in which the number of pressure levels to output exceeded the number of model sigma-z levels. The common “A” array needed to be allocated and used appropriately in the Z dimension depending on its use for sigma levels or pressure levels.

VERSION 6.2.11 – Nov 9, 2019

*Added code to allow IAEROLBC and ICO2LBC = 1 and work correctly for open boundary conditions with LSFLG=0,1. This is essentially set the lateral boundaries for non-sourced aerosols and non-sourced CO2 to prevent these scalars from continually diminishing over time.

With BCTAU set to a timescale less than the long timestep, the boundaries will be set to initial conditions every timestep. This can be quite useful for sulfate aerosols, for example, that do not currently have surface sources. Thus, we need to force the boundaries to keep a source coming in at the boundaries.

- *Added lateral boundary conditions for KPP variables in "rbnd_nonscalar.f90".

- *Modified IKPP to equal 0,1,2 and reduced the number of output diagnostics.

- *Added flag for KPP for UBMN called "UBMN_KPP" which controls the minimum wind speed to use for computing U^* , T^* , R^* , $sflux_u$, $sflux_v$, $sflux_t$, $sflux_r$ for LEAF3, SiB, and KPP wind stress. Need higher UBMN for ocean mixing and SST prediction. Also need more UBMN for getting more correct ocean surface fluxes. Be careful with the value of UBMN_KPP.

- *Set KPP ocean turbidity Jerlov value to 1 for clearer ocean. Also changed default ocean homogeneous profiles to be related to a publication by Emily Riley-Dellaripa.

- *Changed to using LEAF3 surface flux momentum fields rather than separately calculated wind stresses in KPP. These are essentially identical anyways and can be simply passed from LEAF to KPP.

- *Fixed KPP stored input variables fields that are saved between model timesteps, used at KPP timesteps, and then reset. This allows for intermittent KPP timesteps that can be run at similar scales to the atmospheric radiation routines.

- *Fixed problem with REVU variable "reflect_all" which is a first guess reflectivity variable. It provides a reasonable but not best reflectivity estimate. Better to use a radar simulator, but this gives an idea of the reflectivity field. The problem had to do with uninitialized variables. Some compilers will put zeros in such fields, but others put bogus large or tiny number which cause a problem with the routine if not all hydrometeors are 2-moment. The problem mostly existed with PGF90 compiler.

- *Fixed problem in "DP" file creation with 1000mb height fields from ERA5 or other reanalysis data that give initial heights in Geopotential. We first have to convert to Geopotential-height. The problem was that geopotential at 1000mb can be largely negative. Before the variable conversion, the values can be less than -999, which is the missing data values in the code. So, we convert the data first, and then do the -999 check. Same goes in "isan/astp.f90" for -999 or -998. Be careful of these missing data or bad data assumptions.

- *Major changes to "dgrib1_main.f90", "dgrib2_main.f90", "griber_grb1.c", and "griber_grb2.c". Changed the numbering convention for different gridded data to ingest and updated the names of different variables associated with the different gridded data. Also added ingest of ERA5 specific humidity since ERA5 RH uses a merged RH project that varies from RH w.r.t water to RH w.r.t. ice depending on the temperature. RAMS cannot manage a changing definition of RH. Note that the RAMSIN namelist variable IDATAIN needs to be set appropriately depending on whether your "DP" dataprep files contain RH or specific humidity.

- *Fix to computation of net surface roughness for surface aerosol deposition in "aero_deposit.f90". Using previously computed patch roughness instead of sum of vegetation and soil roughness.

- *Minor bug fix in "mic_misc.f90" for assigning pristine ice number concentration to local variable in microphysics.

- *Added code from Sean Freeman to allow RAMS to be compiled and run with Single or Double precision. To change from one to the other, this requires choosing appropriate flagged section in "include.mk" and running the script "prep_precision.sh" in the RAMS root directory for your choice of single or double precision. Note that double precision will use lots more memory, time,

and output file space. Only use this if deemed necessary. This involves changes to a number of “.c” files and include files for variable type declarations (see block/hdf5_f2c.c, include/utills_sub_names.h, lib/hdf5_f2c.c, lib/parlib.c, mpi/mpass_cyclic.f90, revu/rhdf5_f2c.c). *Fix in REVU for text output type to allow incremental grid point output, such as every other grid point and from only points in a portion of a domain. The former version did not extract sub-domain areas correctly.

VERSION 6.2.12 – May 12, 2020 (SCM version stems from here)

Many of the changes below are made to accomodate the stand-alone, single-column model (SCM) version of RAMS microphysics, aerosols, and radation that stems from this version of RAMS. All future changes to these 3 parameterizations need to make their way into the SCM for consistency. Note that changes for the SCM version can be searched with "SaleebySCM".

*Added CIFNP to the aerosol lateral boundary condition option for IAEROLBC=1.

*Created new “trsets_ns” for setting top and bottom boundary conditions for non-advected scalars (ie. fthrd, bext, pcpr, etc) so these grid points do not have bogus values.

*Moved the top and bottom BCs for “fthrd” and “bext” from the “rad_bcond” subroutine to the new “trsets_ns” subroutine. That way, the calls from “non_scalar_bc” only handle lateral boundaries and not top and bottom.

*Added aerosol optical depth to radiation output. This is variable “aodt” and is the AOD in radiation band-3 which is the visible band (~300-700nm). Added “aodt” to “rad_bcond” for lateral boundary setting. Note that this is not single wavelength AOD. You would need to use offline radiative transfer tools to get wavelength at 550nm for example.

*Reordered some variables in “rthrm.f90” for matching with SCM version.

*In “rtimh.f90” subroutine “timestep”, (1) moved “call aerosols” to occur right before “call micro”, (2) added extra “call thermo” after 1st “call negadj1” to update “rv” and “theta” if microphysics variables were altered in the negative adjustment scheme, (3) moved up “call trsets” and added new “call trsets_ns” to right after “call micro”, (4) moved “call checkmicro” and “call mckpp_ocean_mixing” to after the last “call thermo” that occurs right before “call latbnd”. This reordering was done to best match the order needed for the SCM version. Differences in solutions should only be minor.

*Added RAMSIN and code option for freeslip (IFREESLIP=1) lower boundary conditions in which SFLUX terms are set to zero in “sfc_driver.f90”.

*Removed RAMSIN namelist variables “MKCOLTAB” and “COLTABFN” and removed the reading and writing of the microphysics main collection table (mic_init.f90). All the variables for the table are still computed and stored in memory as usual; we are just not writing out the file which can be problematic for parallel computing situations and migration to the SCM version. This change does not impact any model results.

*Added RAMSIN and code option for single column model mode (ISCM==1) where some modifications are made to match up the full RAMS version to the SCM version and output some column data for driving the SCM.

*Modifications to aerosol deposition (aero_deposit.f90) for running in SCM mode in which some land surface information is not available and default values are needed. Differences appear for ISCM=1.

*Modifications to “aero_sources.f90” for dust and sea-salt. Differences should not alter model solutions unless ISCM=1 for single column mode in which some default surface type

information much be applied. The order of passed variables in routine calls were modified so as to easily comment out some variables in the SCM version.

- *Some code reformatting for “check_micro” and “mic_aero.f90”. Moved “check_micro” to it own f90 file. Also moved “negadj1” and “adj1” to its own f90 file (mic_adj.f90).

- *Reformatted some code in “mic_driv.f90” and moved some constants to “micphys.f90” for better organization.

- *Updates to “write_scm” for additional I/O.

- *Modified how we compute extra radiation levels in “rad_driv.f90” and “rad_mclat.f90”. It was formerly done by using reference state PI0 values, which will not be available from other parent models for the SCM version. So, the dependence on PI0 was removed and replaced with model top height and pressure estimates from the McLatchy lookup tables. This appears to provide nearly identical results in all my test cases.

- *Altered computation of “exner” in radiation to remove dependence on PI and use pressure instead. Makes only precision differences in solution due to different arithmetic.

- *Fixes to the RAMS-HUCM interface related to memory allocation errors. This involved removing the passing of 3D micro arrays and using the module “use” statements instead so as to avoid segmentation faults when not all micro variables are needed and allocated. Also a fix to a couple of “if” statements related to graupel and hail in HUCM that were not closed. Finally, removed declaration of large 3D/4D arrays and used the allocated/deallocate commands to avoid segmentation faults resulting from too much stack memory demand.

- *Change to the ISAN package to recompute the moisture variable after the final hydrostatic integration. This helps avoid a mis-match between the varfile moisture and the original moisture form the reanalysis used to initialize and nudge the model. Added a lot of ISAN comments after code tracing this, so that it is easier to identify what is going on in the code.

VERSION 6.2.13 – July 24, 2020

- *Lots of modifications here for the single column model (SCM).

- *Added some simplifications in rthrm.f90 in the call and implementation of “subroutine wettrm”.

- *Added “readwrite” call for the SCM in rtimh.f90. Modified updating of radiation temperature tendency to a new routine within the physics calls associated with the SCM.

- *Add calls to “subroutine thermo” to keep “theta” and “rv” in sync with “thp” and “rtp” for the purposes of the SCM.

- *Created “subroutine predthp” to rtimi.f90 for updating radiation tendency separately from the other scalars for the purpose of the SCM.

- *Added Sean Freeman’s new way of defining single vs double precision run via modifications to utils_sub_names.h.

- *Added SCM flags to RAMSIN to tell RAMS which I,J column and time to output the text files needed as input to the SCM.

- *Created more contained memory allocation of the Level=3 microphysics sedimentation arrays, and modified calls to subroutines in mem_micro.f90 and alloc.f90.

- *Minor call modifications to “subroutine enemb” and sedimentation table allocation in mic_driv.f90.

- *Minor changes in mic_init.f90 for how to compute “tair” for initialization of hydrometeor internal energies in “subroutine initqin”.

- *Modifications to para_init.f90 to allow output of a single columns for the SCM when running a simulation in parallel, and identify which node contains the column in question.

*In rad_driv.f90: (1) Changed from use of THT tendency to FRTHRDP scalar as the radiative temperature tendency to update within the calls to the physics for the SCM. (2) Simplified calls to “radcomp3”. (3) Removed subroutines “cloud_prep” and “rad_enemb” which are obsolete.

VERSION 6.2.14 – August 17, 2020

*Modified calls to “subroutine thermo” for consistency and simplified some variable declarations.

*Modifications to “subroutine timestep” in rtimh.f90 to simplify and accommodate the SCM.

*Eliminated “subroutine initqin” which initialized q2,q6,q7. This was removed since q2,q6,q7 are now treated as advected scalars and are initialized as zero by default. The initialization to non-zero quantities was often overwritten anyway by copybacks in the microphysics driver. Tests show that using zeroes initialization has minimal affect on model solutions. Making these variables advected scalars keeps the hydrometeor columns in sync with internal energy columns. Verified this treatment with Bob Walko. They added this method to OLAM a while back.

*Removed “call azero” for “WP” in rinit.f90 since it seemed redundant given that all 3D scalars are zeroed out at initialization. Tests indicate that this call was irrelevant.

*Tied the ISCM namelist variable to the number of the grid that is to be targeted for outputting a column for the SCM. Ngrid=1 is the parents grid, ngrid=2 is the first nested grid, etc.

*Modified dust lofting so that all methods use the Fecan soil moisture / soil type limitation for lofting based on soil wetness associated with given soil types. This cuts down on lofting when using the Ginoux/Walker type schemes that use erodible fraction databases.

*Eliminated use of “rtgm” in dust lofting and will be using “rtgt” instead since lofting is done on the scalar T grid. This is a minor bug fix. Also eliminated use of “rtgm” in “subroutine get_u10_v10” which estimates the 10m wind for the dust and salt lofting schemes. Made these changes so that the SCM only needs “rtgt” and “topt” for scalar point columns.

*Updated mic_adj.f90 to add “q2,q6,q7” to the negative adjustments in associations with changes to hydrometeor mixing ratios.

*Updates for readwrite routine for the SCM, and update para_init.f90 for SCM parallel I/O.

VERSION 6.2.15 – October 6, 2020

*Removed subroutine “therm_bcond”. Do not need this since adding call the “thermo” after microphysics is run. This automatically updates the boundary conditions for “theta” and “rv”. Needed to do this to eliminate discrepancies in the “RTP-RV” diagnostic for total condensate due to differences in updating between prognostic RTP and diagnostic RV.

*Turned off the monotonic flux limiter advection in the vertical (IMONO=0). Bugs were found in this routine and it needs more testing. Reverting to the default vertical advection routine “fa_zc”.

*Set up code to allow us to easily change the minimum hydrometeor mixing ratio permitted for doing microphysics calculations. We typically set this at 1.e-12 kg/kg. RAMS from ATMET switched to 1.e-9 kg/kg. This should undergo more testing, but for now this is defaulted to 1.e-12 kg/kg as usual.

*Allowing cloud droplet to sedimentate now. Changed the do loop call to sedimentation in mic_driv.f90 to go from 1 to 8 for all hydrometeor species. In the past it only went from 2 to 8 and excluded cloud droplets because their fall speed is so slow. However, recent tests have shown that we must allow droplets to fall in order to simulate fog. This should be tested further in other environments.

*Moved minimum mean diameter (dmb0) for rain, aggregates, and graupel back from 0.3mm to 0.1mm. We had changed these to 0.3mm in some previous versions since it helped alleviate some cloud/stable layers, but now that we have resolved the artificial layering issue (as discussed below) it is best to set dmb0=0.1mm.

*Removed an obsolete routine in radiation.

*Fix in radiation use of GNU via Adele Igel. She found that the radiation lookup tables only use GNU=1 or 2. So there was a disconnect between the microphysics GNU and radiation GNU. Her solution was to use effective diameter (which is less sensitive to GNU, see Harrington's thesis) to compute an updated characteristic diameter, Dn. Then use this Dn and limit the use of radiation GNU to 1 or 2. See the code for detail in subroutine "cloud_opt".

*Changed back an alteration made in RAMS 6.2.08 in which I moved the addition of surface fluxes to after the diffusion is applied to heat, moisture, and momentum. The previous method applied the fluxes to the variables first and then applied diffusion tendencies. I had changed this with the thinking that this might provide better RCE balance; however, I think the original method may allow for better mixing of surface fluxes in to the boundary layer. This could use more testing.

*Notes on IWETEN flag in turb_k.f90 "subroutine bruvais":

1. Using IWETEN=0 computes a moist-Brunt-Vaisala frequency (MBV) using Theta-V and total water mixing ratio as discussed in Durran & Klemp (1982), hereafter DK82, which is applied to all grid cells.

2. Using IWETEN=1 computes the MBV but applies it to only grid cells WITHOUT cloud water. For grid cells with cloud water, a saturated-BV (SBV) is computed from DK82 and is applied to all grid cells WITH cloud water ($RC > 0$).

3. The final computed BV is used in the computation of the Richardson number in the diffusion "subroutine mxdefm". If Ri is deemed unstable, we get mixing. It has been found that using the SBV around clouds often leads to unstable conditions and we get more vertical diffusion near cloud layers. This seems to work well for higher shear conditions and when simulating things like fog, stratus, and stratocumulus. In lower shear conditions and perhaps with coarser DZ, the use of SBV for $RC > 0$, sometime leads to generation and maintenance of unrealistic cloud layers. Following much testing, I suspect the persistence of layers with IWETEN=1 and 2M microphysics is linked to slower evaporation rates of hydrometeors, which is linked to the size of the particles, which is not explicitly set in the 2M microphysics.

4. We have decided to keep IWETEN=1, keep the $RC > 0$ condition for using SBV, but add another constraint for using SBV when:

RV (water vapor mixing ratio) $>$ $FRACSAT * RVLS$ (saturated water vapor mixing ratio).

FRACSAT is a new user-defined value in the RAMSIN namelist. By setting FRACSAT=1.0, you require saturated conditions be met at a grid cell with cloud water ($RC > 0$) in order to use the SBV. If $FRACSAT < 1.0$ and $RC > 0$, then there is the situation of applying SBV when cloud water exists, but conditions are sub-saturated; this could occur along cloud edges where evaporation should be occurring.

5. In tests with bad anomalous cloud layering, the use of FRACSAT between 0.96-0.99 tends to eliminate the layer since SBV is not applied at cloud grid cells that are slightly sub-saturated.

VERSION 6.2.16 – October 14, 2020

*Via Sean Freeman, we have updated HDF5 I/O routines for upgrading from hdf5-1.8.9 to hdf5-1.10.x. Only newer hdf5 versions are available on some supercomputing systems, so the upgrade was necessary.

*New flag for using hdf5 with parallel compression. This feature can be turned on in the “include.mk” compile file. At this time, this should be considered experimental because sometimes parallel compression simply does not work. If it fails at runtime, it will fail at the first model analysis file write time. So you can try it out and know right away if it will function or not.

*Sean Freeman also updated the hdf5 file/variable chunking size determination for better optimization.

VERSION 6.3.00 – October 28, 2020

*Removing some unused variables in the calls to “predict” from the subroutine “hadvance”. Also tested out some preliminary code there to install the R-A-W filter. See notes in the code about installation. Testing shows that there are some issues in the installation that need a deeper look.

*In micphys.f90, changed the minimum hydrometeor mixing ratio allowed from 1.e-12 to 1.e-9 kg/kg. This value is used for positive definite application for hydrometeor mixing ratios and for determining if a subroutine is run within the microphysics. This prevents wasting runtime or nearly negligible amounts of condensate. Testing show that this threshold change has only minor impacts to solutions since these amounts are still quite small. However, runtime is sped up, and compressed file space is reduced.

*Changed “r3min” and “rimin” within subroutine “make_autotab” from 0.01e-6 to 0.01e-8g/cm3. This allows for lookup table space involving collisions between rain or ice with cloud water. This change prevents very small amounts of collector mass from collecting cloud water at a more rapid pace. In a stratocumulus test, this reduced the frequency of occurrence of small rain rates. This should be more realistic.

*Small change in land surface model to prevent read-in soil moisture from exceeding the volumetric capacity that coincides with the LEAF/SiB soil classes.

VERSION 6.3.01 – December 1, 2020

*Removing some unused / obsolete subroutines. Standardizing syntax related to newer subroutines and calls.

*Bug fix to HUCM-SBM microphysics that was only recently accidentally introduced due to a merging of minimum hydrometeor mixing ratio thresholds between the bulk and bin microphysics. This was fixed to keep thresholds separate in the separate microphysics schemes.

*Additional fixes to HUCM for aerosol tracking and microphysical budgets related to vapor growth of cloud and rain.

VERSION 6.3.02 – August 5, 2021

*Added smoke aerosols, that is, absorbing carbon (AbCarb,ABC) aerosols with 2 modes. Absorbing carbon (ABC) is defined here as a mixture of organic carbon (OC) and black carbon (BC). We chose AbCarb-1 as 1%BC, and AbCarb-2 as 2%BC. The AbCarb modes are added as aerosol species numbers 8 and 9. Regenerated aerosols are moved to species 10 and 11, or more generally as “aerocat” and “aerocat-1” where aerocat = the number of aerosol categories. So the regenerated species will always be the final two categories from now on.

*Smoke aerosols are added to microphysics for nucleation and to the radiation routines. New radiation lookup tables were added for brown carbon based on what is used in WRF-chem. But we chose to represent them as 1% BC and 2% BC. More BC = more absorption. We also altered the sulfate radiation tables to be more scattering and less absorbing. Sulfate is now assumed to be 80% soluble and 20% insoluble dust core. The dust in the core has assumed imaginary index of refraction of 0.0015i for UV and visible wavelengths. The former dust core assumed $R(im)$ of 0.008i, which we considered too absorbing.

*Updates to RAMSIN to add IABCARB for turning smoke on and off. Also added ABC1_MAX and ABC2_MAX to input maximum smoke number concentrations for initializing smoke aerosols in the two modes. The chemistry and size of smoke are in places 8 and 9 in IAERO_CHEM, AERO_EPSILON, AERO_MEDRAD. Added initialization routines for smoke in mic_init.f90 named “init_absorbing_carbon” and added the call to this in rdint.f90.

*Bug fix to tracking of aerosol mass in aerosol deposition. Needed to add a density factor.

*In grid_dims.f90, updated “maxisn”, “maxtimes”, and “maxsigz” from 100 to 200 to consider the use of potentially more varfile vertical levels to store and more file times to create.

*Update to mic_adj.f90 to restore aerosols when small values of hydrometeors are set to zero. This helps prevent too much artificial removal of aerosols that are contained within hydrometeors, but rather restore this mass and assumed number to the regenerated aerosol species.

*Added a function to keep sedimentation table parameter “densrtgt” from getting too large if the model top density gets very low for model simulation that go very high up. This was occurring for the RAMS SCM version applied to CCPP for host model tests with model tops over 50km AGL.

*Modifications to aerosol radiation in rad_aero.f90 to add absorbing carbon / smoke and also changes to the dust modes to include dust that is low-absorbing, mid-absorbing, and high absorbing. The associated imaginary indices of refraction (UV and VIS wavelengths) were 0.0015i, 0.003i, and 0.008i, respectively. Past versions used 0.008i which is typically too high. The default choice can be chosen in rrad3.f90 with the hardcoded flag “dust_ref_im”; it will be set as “2” by default which uses the mid-absorbing values of 0.003i. This can be user altered depending on the type of dust absorption you want to simulate. If you make a change to “dust_ref_im” you need to do a full model recompile for the change to take effect.

*Small bounds checking change in the aerosol deposition code.

*Set all “A80” string limits related to namelists to “A256”. This limit was causing random segmentation faults if users used really long strings in the RAMSIN namelist.

VERSION 6.3.03 – March 2, 2023

*Set IMONO=1 in radvc.f90 to turn the monotonic flux limiter back on. We should still check this to see if there are bugs in this routine related to density and pressure. But some results show this performs better along boundaries and helps prevent aerosol concentrations from getting too high in unusual circumstances.

*In the BULK microphysics only, I separated out microphysics process rate variables for condensation/deposition and evaporation/sublimation. So, for example, VAPCLDT is now the vapor deposition growth only for cloud water categories (+ values only). EVAPCLDT is the evaporation of the cloud water category only (+ values only). In past RAMS versions, VAPCLDT was the net of vapor growth and evaporation summed over time between output writes (+ or – values with + being net growth and – being net loss). The new evaporation process

rate is given a positive value now, not negative. Also added process rates FREEZINGT and MELTINGT that are totals computed in association with the computation of the latent heating rates that are associated with net phase changes in the microphysics. NOTE THAT THESE CHANGES ONLY APPLY TO THE BULK MICROPHYSICS AND NOT THE HUCM BIN MICROPHYSICS.

*Eliminated the RAMS version directory below /src so that we do not have to keep changing the version directory name on Github. This is also reflected in the “include.mk” file by removing the “RAMS_VERSION” flag since it’s no longer needed.

VERSION 6.3.03a (temporary) – April, 2023

*Updated lateral boundary conditions for nested grids so that features translate from parent grid to nested grid more accurately and seamlessly. There are some hard coded coefficients in bc/rbnd.f90 and/or nesting/nest_intrp.f90 that could be customized as needed, but default values are reasonable for most situations. (Work done by Leah Grant)

VERSION 6.3.03b (temporary) – June, 2023

*In microphysics, put a cloud droplet nucleation limiter at 0.01microns (10nm) so that if the aerosol distribution median radius drops below 10nm, nucleation is stopped. Our nucleation lookup tables do not go below 10nm median radius. In the past, nucleation could continue as the median radius of the aerosol distribution was forced to the minimum size. In some situations it seemed that overnucleation was occurring, and a limiter was perhaps needed.

*Nucleate cloud droplets at the minimum droplet size instead of diagnosing a size based on nucleation number and available vapor. This will reduce the amount of mass in the microphysics budgets for nucleation, but this is compensated for via the vapor condensation routines.

VERSION 6.3.03c (temporary) – July, 2023

In microphysics, updates were done to permit better prediction of heterogeneous ice nucleation when values are relatively small. Ryan Patnaude, (Kreidenweis & DeMott group) used INP data and ice crystal concentration data from the SPICULE field project from congestus clouds for initializing and validating RAMS ice nucleation scheme. The following code changes greatly improved the prediction:

*In mic_adj.f90, I modified a condition statement from $cx(k,lcatt) \leq 0.0$ to $cx(k,lcatt) < cxmin$ to check cx and rx and set things to zero if needed. Former method occasionally created NaNs due to attempted division by zero.

*Add call to “psxfer” in the mic_driv.f90 after the call to “colxfers”. This is necessary so that cloud ice and snow can be moved to the correct category after collision-coalescence. Otherwise, the number adjustment routine “enemb” can artificially add pristine ice number concentration in order to keep the mean pristine ice diameter in bounds from 15-125 microns.

*In mic_nuc.f90 and mic_nucpre.f90 added an IIFN=2 constraint so that the DeMott formulas are only applied between temperatures of -35C to 0.0C. In the past we permitted this to occur at colder temperatures, which perhaps should not be done since the formula is not meant to be extrapolated below -35C.

*In mic_nuc.f90 heterogeneous ice nucleation, I modified the nucleation limiter code from “diagnc-cx(k,3)” to “diagnc-cx(k,3)-cx(k,4)” to also consider the number concentration of snow

in addition to pristine ice since snow is just vapor grown pristine ice. This seems to help limit the heterogeneous ice nucleation. This provides more reasonable concentration compared to SPICULE observations mentioned above in this section header.

*In micphys.f90 change rxmin to 1.e-16 and cxmin to 1.e-5 to permit lower pristine ice concentrations that occur from INP at warmer temperatures.

VERSION 6.3.03d (temporary) – Aug, 2023

*In mic_nuc.f90 heterogeneous ice nucleation section, decided to permit nucleation at temperatures less than -35C, but only allow the DeMott formulas to provide an air temperature down to -35C. So, if actual air temperature is -45C, the DeMott formula will use a max cold temperature of -35C. This should permit nucleation without over extrapolating the DeMott formulas.

VERSION 6.3.03e (temporary) – Aug, 2023

*Throughout the RAMS code, I changed the CCN and GCCN naming conventions since GCCN do not actually have to be giant. They can be whatever size and solubility we choose. So, providing more generalized names avoids the issue of thinking we can only specify giant particles in the GCCN variable. The 2 dust modes, for example, are named MD1NP and MD2NP for the mode 1 and mode 2 number concentrations. They could be any size we choose. So, I've changed CCN and GCCN number to CN1NP and CN2NP. I've also changed their mass to CN1MP and CN2MP. This way we could have 2 aerosol modes that are Aitken mode and accumulation mode or say, accumulation mode and ultra-giant mode. There's flexibility here as to what the modes of aerosols can be.

VERSION 6.3.03f (temporary) – Aug, 2023

*Added optional IFN(ice forming nuclei) / INP(ice nucleating particles) initialization option in the subroutine "init_ifn" in file mic_init.f90 that allows the user to initialize simulations with the:

1. SPICULE field campaign fitted vertical profile from aircraft obs assuming the user sets IIFN=2, IFN_FORMULA=2, and CIN_MAX = -1.0 (or)
2. MC3E field campaign fitted vertical profiles from aircraft obs assuming the user sets IIFN=2, IFN_FORMULA=1, and CIN_MAX = -2.0.

These are reasonable continental profiles and likely a better default choice than the former default exponentially decreasing profile with height. When using the exponential profile (for CIN_MAX > 0.0, it's probably most realistic to get CIN_MAX >= 0.5 /mg. Smaller values are probably too small even for a default scenario.

VERSION 6.3.04 – Aug, 2023

*Combined all the 6.3.03x subversions into this combined version. Updated the documentation as well as the sample RAMSIN test simulations in the bin.rams directory.

*Added RAMSIN namelist flags ISPONG_PTS and SPONGE_TAU related to the nested grid sponge layer which works to better transition scalar from a parent grid to nested grid.

NOTE THAT ALL FUTURE UPDATES WILL BE ADDED TO THE GITHUB LOGS BUT WE WILL KEEP A SUMMARY OF CHANGES HERE FOR EASE OF READING.

VERSION CHANGES v6.3.04 to v6.3.04a – Feb 21, 2024

- *Updated ice aggregation efficiencies in bulk microphysics.
- *Array size fix for “concen_tab” in subroutine “cldnuc”.
- *Some updates to Makefiles

VERSION CHANGES v6.3.04a to v6.3.04b – May 1, 2025

- *Added a third CCN mode “cn3np”, “cn3mp” scalars and CCN3_MAX in namelist.
- *Updates to default Lite file truncation values for some variables.
- *Updated limits for haze nucleation in bulk ice nucleation routine.
- *Updated cloud droplet nucleation routines and removed the irrelevant saturation lookup tables.
- *Added ITRUNCLITE and IKERNELA to the namelist.
- *Major update to cloud droplet nucleation lookup tables to drastically improve droplet nucleation in the presence of pre-existing cloud droplets and for RAMSIN option ICCNLEV = 1,2.
- *Update aerosol lateral boundary nudging to 2 interior grid cells.
- *Fixed point of problem for too many restored aerosols when small hydrometeor mixing ratios and numbers are set to zero (mic_adj.f90).

VERSION CHANGES v6.3.04b to v6.3.04c – July 11, 2025

- *Removed all of REVU. We consider this obsolete. It could still have use for showing examples of how to compute post-processed variables. As such, access this from earlier GitHub versions.
- *Changed the names of many microphysical process rate variables and added many new rate variables as well. See the RAMS variable document list for details on exactly what each rate output includes. Some process rates were updated to be more accurate such that those from collision/coalescence that can be adjusted in subroutine “colxfers”. This adjustment was not considered in earlier versions, but was a reasonable estimate.
- *Adjustment of minimum aerosol size in mic_adj.f90 and mic_nucpre.f90.
- *Bug fix in mic_coll.f90 related to number self-transfer as “mx” and “jcat” were misused.
- *Adjustment to ice nucleation process rate diagnostics to include max limiter as is done for the inclusion of new ice particles for contact nucleation and homogeneous freezing nucleation.
- *Bug fix in leaf3_soilveg.f90 subroutine “grndvap”. Removed the “alpha” term from variable “ground_rvap”. This changed was investigated and verified by Aryeh Drager.
- *Change in therm_lib.f90 subroutine “qtc”. Altered computation of “tempc” to change value “80.” to the more precise “79.79” in case this causing increasing problems over time. This changed was investigated and verified by Aryeh Drager and Bob Walko.

VERSION CHANGES v6.3.04c to v6.3.04d – December 11, 2025

- *Add some aerosol tracking variables to the lite file truncation.
- *Add code to mic_adj.f90 to zero out “aerosol mass in hydrometeors’ within 2 grid cells of the lateral boundaries on the outermost domain in order to avoid some potential advection and diffusion issues that sometimes arise along open boundaries with topography. This is NOT done for periodic boundary conditons.
- *Reinstituted minimum cloud droplet mean mass required to initiate hybrid-bin riming in order to keep with LUT minimum. A similar minimum is applied for bulk riming.

VERSION CHANGES v6.3.04d to v6.3.05 – August 31, 2026

*This jump in versions combines many separate developments made by Steve Saleeby, Adele Igel, and Bee Leung. Saleeby kept updating sub-version tags in his commits to his RAMS fork. We first merged the version from Adele & Bee in to the RAMS main repository version v6.3.04 and then merged Saleeby version tagged as v6.3.04d onto this. Named this new version v6.3.05.

*The version from Adele/Bee contained updated for large scale forcing, the addition of BUGSRAD radiation and RTE+RRTMGP radiation, ZPF compression (from the INCUS version of RAMS), a few surface flux modification, and few other updates that can be found in the logs.

*Notes on BUGSRAD:

a. Adele added BUGSRAD radiation as option 4 for ISWRTYP and ILWRTYP. A few notes about this implementation:

b. Only cloud droplets, pristine ice and snow are passed to the BUGSRAD radiation routines. This is ok for now as cloud droplets and ice crystals are the two most important hydrometeors for radiation.

c. Cloud droplet mass and number are both passed. Only the sum of pristine ice and snow mass is passed.

d. No aerosol species are coupled to BUGSRAD radiation.

e. BUGSRAD contains separate albedo values for direct and diffuse radiation. I've set these to be equal. (see radiate/driver_read.f90, alndr, alndf, alvdr, alvdf).

f. BUGSRAD contains separate albedo values for shortwave and near IR. I've set these to be equal. (aln* - near IR, alv* - visible)

g. BUGSRAD has its own precision definitions (e.g. dbl_kind). They're still being used and aren't interfaced with the options to compile with double or single precision in the Makefile.

h. Also added RTE+RRTMGP radiation (<https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2019MS001621>) as option 5

*Adele Igel's changes as of June 2026:

(a) ZDELAY: Height to begin vertical grid stretching. Useful for keeping constant grid spacing in the boundary layer and stretching above the boundary layer top.

(b) ISFCL = 3: Use this when you want to specify everything about the surface simply: Momentum fluxes are hard-coded to 0. Latent and sensible heat fluxes are given in W/m² by DRTCON and DTHCON, respectively. Surface albedo is specified by ALBEDO. Surface temperature is specified by SEATMP and is used only for calculating RLONGUP. When using this option, only the TOPT file will be made. The surface, NDVI, and SST files are unnecessary. Also, using this scheme, patch properties such as ustar are not calculated. As such, the aerosol deposition scheme cannot be run. A fatal error has been added if a user tries to use the deposition scheme with simple surface fluxes.

(c) ZTROUGH and ZMROUGH: ZTROUGH is used for all scalars, ZMROUGH for winds. These values are only used for soil patches when ISFCL = 0 or for the water patch if ICHARNOCK = 0. The LEAF and SiB schemes continue to use only one value of roughness.

(d) ICHARNOCK: Flag to use the Charnock ocean roughness parameterization. If set to 0, instead use ZTROUGH and ZMROUGH for the ocean patch.

(e) DIVLS: Set large scale divergence rate. A subsidence profile is derived from the DIVLS value where the subsidence at the surface is always 0 m/s.

(f) IIFN = 4: Use quasi-constant pristine ice + snow number concentration. Still need IPRIS and ISNOW = 5. Quasi-constant value is given by CIN_MAX in # per LITER!!! This

differs from the other uses of CIN_MAX, but is consistent with typical specifications for ice crystal concentration in the literature.

(f) IUGFORCE: If =0, off; if=1, use sounding profile winds, if=2, use forcingfile winds that vary in time.

(g) FORCINGFILE: File name that provides geostrophic forcing wind and/or simple 0D time-varying SST values. Uses the DEPHY file format.

(h) IUPDSST = 2: Update SST from FORCINGFILE in time. Uses a single value for the entire domain.

(i) IO3FLG: If =0, ozone profile given from McLatchy, if =1, ozone profile included in RAMSIN namelist or SOUND_IN in ppb

(j) O3S: Input ozone profile, same length as other input sounding profiles

(k) SOUND_FILE: Input sounding profile name, typically SOUND_IN

(l) Surface/Radiation: For ISFCL = 0 and ISFCL = 3, changed surface longwave emissivity to 0.98 to make it consistent with the bare surface/water emissivity used by LEAF-3.

(m) Radiation: For INITIAL = 1 or INITORIG = 1, radiation will now use the full input sounding where the input sounding extends beyond the prognostic model top. McLatchy profiles will fill in only above the input sounding, if necessary. To facilitate this change, there is a new subroutine in rad_driver.f90 called prep_atm_profiles. It contains iaction=3 call to mclatchy and the block of code to check for negative values. radcalc3, radcalc4, and radcalc5 all call prep_atm_profiles.

(n) Radiation: added aerosol optical properties to the RTE-RRTMGP scheme.

(o) Thought about surface/radiation: Currently, sfcrad calculates the total upward longwave radiation – this is predominantly emission from the surface, but includes a small portion that is reflected longwave radiation. The reflected portion is calculated using the previous radiation calculation's LWDN. Can we not have the surface scheme only calculate the emitted portion and separately calculate a longwave average emissivity (albedo)?

* Bee Leung's changes as of June 2026:

(a) Add radiation diagnostic variables: RLONTOP: outgoing LW at top of atmosphere, FTHRDSW: radiative heating rate SW component, FTHRDW: radiative heating rate LW component

(b) Update CO2 concentrations used in radiation options 3-5. Updated this for RCEMIP. Previously CO2 concentration was hard coded to be ~380ppm in Harrington radiation in Subroutine path_lengths. These concentrations have not been realistic for some decades now ☹ Changed it to 420ppm for Harrington, BUGSRAD, and RTE-RRTMGP. Right now it is hardcoded at compilation, could make this a namelist option in the future.

(c) Update IDIFFPERTS=1. Previously, IDIFFPERTS=1 used a different equation for THP vs. other scalars. Discussed with Leah Grant that there doesn't seem to be any reason to do this. Changed so that IDIFFPERTS=1 uses the same equation for all scalars. This fixes energy balance over very long times in RCE mode.

(d) Merge ZFP compression changes made by Sean Freeman for INCUS development. This improves compression of lite files, with namelist option to set desired accuracy. Requires adding ZFP library and setting HDF5_PLUGIN_PATH, but this is optional at compilation using flag "-DENABLE_ZFP_COMPRESSION".